

BE GREEN HEROES

EDITION **3**



FIGHTING WIDESPREAD
MISINFORMATION AND PURSUING
ENVIRONMENTAL LITERACY

HANDBOOK OF TEXTILE SUSTAINABILITY

SAE CHANG



BE GREEN HEROES

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EDITION 3 – Chapter I

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By Sae Chang

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This chapter represents part of a larger scientific discussion. Completion of the entire book is required to fully understand the context and conclusions represented.

Start Your Journey as Green Hero



Chapter I: Awareness Before Action

The foundation of responsible green decision-making

What this chapter covers

-  How green misinformation spreads
-  Why perception often differs from reality
-  How scientific evidence guides better environmental decisions
-  How practical insights translate into actionable decisions



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العنوان:

أبطال الكوكب الأخضر

مواجهة التضليل ونشر الوعي البيئي

رابط الموقع الإلكتروني ورمز QR

غير متوفر حالياً

Chinese

书名: **成为绿色英雄**: 对抗广泛传播的错误信息，提升
环境素养, 纺织品可持续发展手册

网站链接与二维码:

<https://acesse.one/c7cH7>



English

Title: **BE GREEN HEROES: FIGHTING WIDESPREAD
MISINFORMATION AND PURSUING ENVIRONMENTAL
LITERACY, HANDBOOK OF TEXTILE SUSTAINABILITY**

Website link and QR code:

<https://amzn.to/4oz7yJK>



French

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DÉSINFORMATION GÉNÉRALISÉE ET PROMOUVOIR LA**

LITTÉRATIE ENVIRONNEMENTALE, MANUEL DE
DURABILITÉ TEXTILE

Lien web et code QR:

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DISINFORMAZIONE, PERSEGUIAMO L'EDUCAZIONE
AMBIENTALE, MANUALE DI SOSTENIBILITÀ TESSILE**

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여정, 섬유·의류 지속가능성 핸드북

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DESINFORMACIÓN COMÚN Y LA BÚSQUEDA DE LA
ALFABETIZACIÓN AMBIENTAL, MANUAL DE
SOSTENIBILIDAD TEXTIL**

Enlace web y código QR:

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Edition 3 Notes:

This third edition has been thoughtfully refined to enhance readability for a broad audience, welcoming both casual readers and industry professionals. Passages offering in-depth insights for industry professionals and students in related fields are clearly marked as *****Industry Specifics*****. These sections provide valuable technical detail for those who seek it, while allowing casual readers to bypass them without losing the flow or central emphasis of each topic.

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Chapters 2, 3, 4 and 5 are not included in this distributable edition. To access the complete text of *BE GREEN HEROES*, please refer to the links provided on Pages 6 ~ 7.

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About the Author

My name is Sae Chang. I am a textile engineer with hands-on experience spanning nearly every stage of textile manufacturing, from fiber production to garment assembly. Over the course of my career, I have worked across engineering, product development, quality control, sales, and marketing, gaining a comprehensive understanding of both the technical and commercial dimensions of the industry. My experience includes roles in multinational corporations as well as leading my own ventures.

Over time, I became increasingly concerned about how misinformation shapes public understanding of environmental sustainability. Many widely accepted narratives are built on incomplete or unscientific foundations, often reinforced by the growing reliance on quick, easily accessible information. This issue is particularly evident in discussions around textiles—an everyday, highly visible industry that is frequently used to illustrate environmental impact, yet often misrepresented in the process.

While textiles do contribute meaningfully to environmental challenges, their familiarity can amplify perceived impacts beyond what is scientifically supported. This creates a complex landscape where valid concerns coexist with oversimplified or inaccurate conclusions. Search engines have long contributed to this imbalance by prioritizing visibility over validity, and more recently, the rapid rise of AI has further accelerated this effect.

BE GREEN HEROES was developed in response to these concerns. In this book, I provide a comprehensive

examination of textile materials, processes, use cases, and end-of-life stages, guiding readers to better understand the true scope of environmental impacts. Drawing from my real-world experience, I aim to expose the gaps between scientific reality and commonly accepted narratives, while encouraging a more critical and evidence-based approach to environmental issues. I have taken particular care to present complex science and processes in a way that allows readers to clearly grasp what truly matters in environmental sustainability.

In parallel, I developed the Clean Recycling Initiative™ technology platform, designed to reclaim textile waste with minimal environmental impact. Recognizing its broader potential, I donated the technology to the non-profit organization of the same name, making it accessible to individuals and organizations worldwide.

Building on this work, the concepts and insights from this book are being integrated into the BE GREEN HEROES (BGH) Web3 learning platform, which operates under the framework LEARN. EARN. ACT. Within this system, participants—referred to as Green Heroes—engage with environmental science through accessible learning formats and are rewarded with Web3-based tokens. Through verified learning and earned credits, they can also participate in the governance of BGH operations. Further details are available on the website below.

<https://begreenheroes.com>



About the Book

Throughout this book, I will present many widely popular environmental initiatives and concepts, which are based on a significant lack of appropriate science, misleading the public. To encourage critical thinking, I propose several perspectives as a prelude, carefully selected to stimulate inquisitiveness in readers' minds.

- Widespread misinformation on the Internet
- Self-proclaimed experts educated by the Internet
- Organizations operating with a significant lack of the necessary expertise, knowledge and science
- Importance of science-based education

Importance of Comprehensive Views

In the realm of environmental management, it is crucial to maintain comprehensive views on the overall impacts of an action. For instance, while carbon emissions are undoubtedly a significant concern, they are not the only area that warrants attention. In my view, chemical toxicity is a critical issue that is largely overlooked in the current global sustainability efforts. Additionally, there are many other important factors that are being ignored. Focusing solely on a particular problem while neglecting potentially more impactful, hidden issues is analogous to only caring for one's skin without conducting necessary internal medical examinations. This neglect may lead to severe consequences that are more challenging to address in the future.

To some readers, my assertions may seem exaggerated. However, I can assure you that they are not. I will highlight numerous pressing issues addressing this concern with specific examples throughout the book. Many of them are closely related to our daily lives, making the associated perspectives easy for most readers to understand and digest.

Although this book primarily focuses on the textile goods we consume, I will also draw parallels with other industries to discuss the effectiveness of global sustainability efforts. For instance, global societies are investing trillions of dollars in specific areas such as the transition from the Internal Combustion Engine (ICE) vehicles to the Electric Vehicles (EV). Some readers might think that vehicles are unrelated to textiles and view this subject as out of context. However, they may be surprised to learn how closely the two are interconnected, as components like seatbelts, airbags, and seats are all textile materials. Moreover, the environmental impacts of any industry do not exist in isolation. They all influence our overall sustainability in one way or another.

Misinformation Sources and Causes

Another crucial perspective of this book is to highlight extensive misinformation prevalent on the internet. With recent advancements in the information technology, we increasingly rely on the internet for information. Google has largely supplanted encyclopedias and textbooks for a significant portion of the population. The advent of the ChatGPT and other AI-based platforms has further enhanced an easy access to both accurate information and misinformation.

Although I am not a software engineer and lack detailed knowledge of search engine mechanisms, my decades of experience using these tools have shown that many widely used soft tools can contribute to the spread of misinformation. For instance, search engine results appear to often prioritize popularity over accuracy - This phenomenon will be discussed in greater depth in **Chapter V: Solutions**.

While, in some cases, popularity may align with accurate information, this is often not the case in the realm of sustainability. Misinformation frequently overshadows true scientific data and accurate information on the internet.

One reason for this phenomenon, in my view, is the proliferation of self-proclaimed experts and organizations, operated or influenced by the misinformation generated by themselves. Areas related to the environment, sustainability, ESG, and climate change have become fertile grounds for individuals and organizations to claim their expertise and label themselves as experts without substantial qualifications. Consequently, there has been a myriad of unsubstantiated assertions, even from highly influential authorities – many examples will be presented throughout the book.

The problems created by these actions are detrimental to global sustainability efforts and must be addressed with the utmost priority. The widespread misinformation and erroneous claims impede genuine progress and undermine the integrity of the global sustainability efforts on the environment.

Structure of the book

I decided to take a slightly unconventional approach in organising the contents of this book. Given its focus on presenting science-based information about various sustainability claims and environmental initiatives, a more traditional structure would begin with basic educational contents before moving on to more practical subjects. Initially, I wrote the book in the order of: 1. Basic science of textile materials and manufacturing, 2. Lifecycle assessment, 3. Consumer-related subjects, 4. Misinformation crisis, and finally, 5. Solutions.

However, I chose to reorder the chapters, placing the consumer-related subjects first. This decision was made with the consideration that it might be easier for most readers without related knowledge or educational/industry experience to digest and follow through the contents. I believe this approach will help readers maintain a higher level of engagement by relating their personal experiences as consumers to increasingly scientific information as they progress. To support this objective, I included various *internal* references to related subjects, helping readers connect the dots with relevant scientific concepts in the dedicated discussions of the subject matters throughout the book.

Reference materials

To ensure the **credibility and transparency** of the information presented, this book includes a wide array of external references to **peer-reviewed scientific research and validated sources**. Furthermore, I have also included more accessible resources, such as blogs and YouTube

videos, provided that their scientific accuracy is ensured and if they are deemed helpful for readers to gain related knowledge more effectively.

Rather than confining citations to appendices or endnotes, as is often customary in academic works, I have chosen a more reader-friendly method of citation. With this principle, each reference is **embedded within the relevant section**, accompanied by **clickable links for e-book users** and **scannable QR codes for print readers**. This enables readers to verify information in real time, deepening both trust and understanding.

For those seeking more in-depth knowledge on textiles, particularly in the context of **thermal and winter applications**, which are among the most environmentally impactful due to elevated material and chemical demands, I encourage exploration of my companion book:

COLD WINTER WARM WINTER: Same Weather Different Protection – A Smart Guide to Staying Warm from Everyday Life to Advanced Textile Innovations, available globally via Amazon, including the following link and QR code:

https://www.amazon.com/COLD-WINTER-WARM-PROTECTION-INNOVATIONS/dp/B0FJY5464M/ref=tmm_pap_swat_ch_0?encoding=UTF8&dib_tag=se&dib=eyJ2ljojMSJ9.OlliAeUlMDZ8WTV9oq_fQ.JvAvlMTMFpw0clYUz11TOP2lPHXUBYmRXeCcQ84wdXw&qid=1753916438&sr=8-1



Industry Specifics

While this book is written with the general public as the main audience group, this book also includes **advanced technical discussions** intended for professionals in the textile and apparel industries, as well as for students pursuing specialized studies in related disciplines. These sections are clearly labeled *****Industry Specifics***** to allow general readers to navigate the book without losing the continuity of the main content.

However, I acknowledge the inherent difficulty in neatly dividing material between “general” and “industry” audiences. The complexity of today’s environmental challenges demands that **all stakeholders - consumers, students, and professionals - develop a stronger understanding of key issues**. In this spirit, I have made a **conscious effort to minimize such segmentation**, integrating technical depth in ways that elevate the reader's knowledge regardless of background.

Before delving into the book, I must ask readers to understand that my expressions may be, at times, blunt and direct. I feel that this is necessary due to the urgency for the global society to address significant challenges we face. While many people trust that societal authorities, such as government organizations, affiliates, and some of the largest NGOs on environmental matters, guide us correctly, I have observed that even highly authoritative organizations often make erroneous claims and engage in greenwashing. Throughout this book, I will illustrate numerous examples of such occurrences with candid expressions.

Credits

This book was written with the assistance of several individuals. From the outset, I recognized the risk of being confined by my own educational and professional experiences. Considering that the primary goal is to convey critical messages about environmental management and sustainability to as wide an audience as possible, the language used in the book must match the eye levels of the audience.

To address this, I sought fresh perspectives from the individuals without prior knowledge or experience in the textile industry. Their ability to comprehend the contents serves as a valuable indicator that the broader public will also find it accessible and understandable.

With this goal in mind, I asked the following individuals to partake in the assigned roles:

Paula Cevallos Agreda is currently in the master's program in the Management – Sustainability of HEC, Montreal and conducted research on some contents of the book, including the experiment on the comparison of power consumption between cotton and polyester in home laundry.

Anna Chumbe is MA student in the Comparative Literature program of University of Montreal. Anna worked on the proof reading in English and helped in editing.

Hyeonji Yu is currently a student at Chungbuk National University, South Korea and worked on some graphics presented in the book.

I sincerely thank the intelligent individuals who helped me on this important journey. Without your help, this book would not be what it is.

Now, let us begin our journey to becoming better informed and educated.

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Chapter I

**Consumers,
Be Aware and Beware!**

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Pre-Chapter commentary

Quote of the chapter

“Given the prevalence of greenwashing, even by some of the most trusted organizations globally, the public's overly simplistic approach to various sustainability claims does little to benefit the environment. It is crucial for consumers to take a more critical stance and demand transparent, comprehensive information in order to make informed decisions that truly benefit the environment.”

Understanding the environmental implications of various textile materials, such as cotton and polyester, begins with examining their origins, whether grown in cotton fields or synthesized as synthetic fibers. By reviewing manufacturing processes to becoming the finished products we purchase and tracing their journey during and after consumption, we gain a critical knowledge base for evaluating the true environmental impacts of different material types.

This chapter aims to provide the essential information required for such an analysis, offering readers valuable insights that may influence not only on the sustainability aspects of their mindsets, but also in other areas of their lives, such as making purchasing decisions and deciding on the maintenance practices of their textile goods.

While some scientific details presented in this book may appear challenging, readers should not feel overwhelmed as I made extra efforts to explain the related topics in easily digestible manners so that even those who do not have

backgrounds in science or engineering can grasp the concepts effectively. By completing this book, you will acquire a comprehensive understanding of textile sustainability.

Life from Beginning to End

Textile goods are indispensable components of human life, encompassing all stages from the birth to the death. This intrinsic connection to humanity underscores the textile industry as one of the oldest in existence. A compelling anecdote is the biblical account of Eve, who, feeling ashamed of her nudity after consuming the forbidden fruit, sought to cover herself with a leaf from a tree, symbolically marking the inception of the textile industry.

Humans, in ancient times, used various materials such as animal fur, leather, leaves, etc., to cover their bodies. From an evolutionary perspective, early humans may have utilized natural materials for warmth and protection after progressively losing their body hair during the evolution.

In many cultures, the significance of textiles extends beyond the life on earth. For instance, it is a common practice for individuals to select the garments they wish to be buried in, reflecting their beliefs in a well-prepared afterlife. The elaborate burial practices of the ancient Egyptian Pharaohs, who had meticulously planned to have their bodies wrapped in multiple layers of linen during mummification and placed in the archaeological wonders of Pyramid, illustrate the fundamental desires of humans even after death. These historical and cultural perspectives highlight the longstanding and integral role of textile goods in human civilization, from the ancient times to the present.

Humans as Social Animals

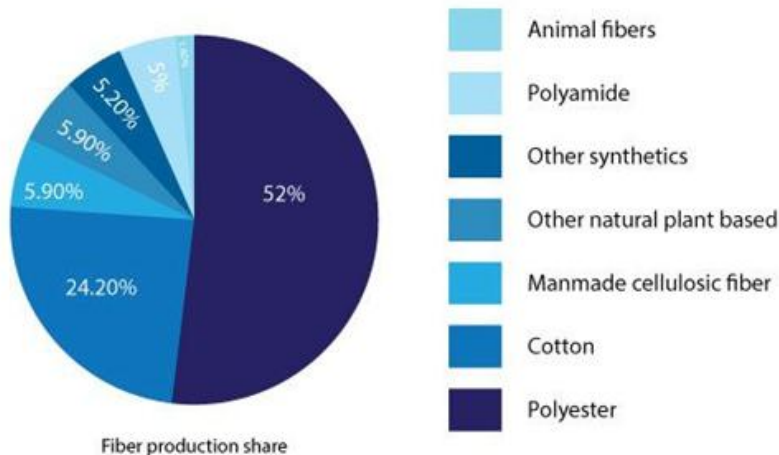
Throughout our lives, from infancy to old age, we consume a vast array of textile goods in significant quantities. The consumption of textile goods is intricately linked to one of the fundamental human instincts associated with social behavior. Anecdotally, we can easily imagine that the biblical figure Eve selected the most visually appealing leaves to cover herself, continuously seeking aesthetically pleasing materials to enhance her appearance for Adam while the later reciprocating such effort. Textile goods have long served as essential tools for self-expressions and social interactions, reflecting our nature as social beings.

Material Consumption Patterns

Most individuals possess a considerable degree of experience and knowledge regarding which types of fibers align best with their lifestyles and values. For instance, I personally prefer cotton for socks and wool for business suits, while I favor synthetic fibers for nearly all other uses. The reasons behind my preferences and purchasing decisions form a significant part of this book, as they are based on comprehensive scientific analysis not only on the part of performance metrics but also on the environmental sustainability. I understand that a large proportion of consumers would prefer to purchase high fashion goods made with natural materials, if financially feasible. This preference may be partly due to a perceived quality and comfort, and partly on environmental concerns over synthetic materials.

To provide a context to this discussion, I present the pie chart of the worldwide fiber consumption in **Fig. 1**.

Fig.1 Worldwide Fiber Consumptions 2020



<https://www.textiletoday.com.bd/global-polyester-fiber-market-share-and-scopes-for-bangladesh>



Dominance of Polyester

Polyester dominates, accounting for 52% of the total fiber consumption. Cotton follows with 24.2%, while all other fibers constitute relatively minor portions. Notably, my personal wardrobe would likely reflect a higher percentage of polyester than the global average.

Some readers may wonder why polyester is so prevalent, especially when many people express a preference for textiles made from natural materials. Various reasons behind this disparity are explored throughout this book, with discussions encompassing the entire lifecycles of key material types from farming to manufacturing, consumption, and disposal. I present a wide range of scientific data, real-world phenomena, and case studies

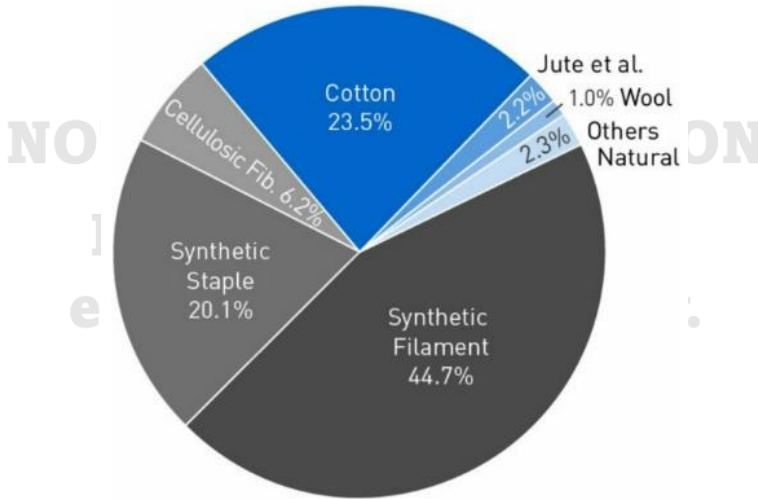
that are closely tied to this topic, enabling readers to naturally observe how this gap is progressively bridged. Ultimately, readers will come to understand that this gap is by no means disadvantageous to humanity, even from the standpoint of environmental protection and sustainability, as they are guided through a series of scientifically grounded explanations that make this conclusion readily accessible.

Given the limited space available, a detailed analysis on every fiber type is not possible in this book. Instead, an emphasis is placed on the two dominant fibers: cotton, representing natural fibers, and polyester, representing synthetic fibers. Nevertheless, these two materials alone account for approximately 75% of global textile consumption. Moreover, the environmental and sustainability impacts of these two fibers are broadly representative of their respective material categories, namely natural fibers and synthetic fibers. In this regard, there is little doubt that they constitute appropriate and effective choices for fulfilling the objectives of this book.

Historical Analysis

In the meantime, historical data analysis reveals that the fiber compositions do not change drastically year over year. A supplementary chart from 2018, with a slightly different categorization of fibers, provides similar splits.

Fig.2 Worldwide Fiber Consumptions 2018



<https://textile-network.com/en/Technical-Textiles/Fasern-Garne/Forecast-world-fibre-production>



For example, in **Fig.1**, cotton constituted 24.2% of the total fiber consumption while it is 23.5% in **Fig.2**. Given that the difference is only 0.7% between 2018 and 2020 and there is an inherent margin of error in these types of statistic figures, this variation is considered negligible. Conversely, the combined share of the synthetic fibers in the 2020 pie chart (**Fig.1**) was 62.2%, comprising polyester (52%) and all other synthetic fibers (10.2%). In 2018, the synthetic fibers made up 64.8% of the market, categorized in the synthetic filament (44.7%) and the staple (20.1%). This small difference of 2.6% on the synthetic fiber consumption can also be viewed as negligible and perhaps within statistical margin of error, indicating that the consumption of the synthetic fibers has also remained relatively stable year over year.

Consumption Rate and Economic Development

The overall quantity of textile consumption is closely tied to economic and technological development. Most textile goods have become more affordable and accessible than ever before, even when considering inflation and shifts in monetary values. This trend is particularly evident for the textile products produced through industrialized manufacturing systems. For instance, the cost of producing a common cotton shirt has decreased significantly over time, thanks to numerous advancements in farming techniques, manufacturing efficiency, and distribution methods, particularly since the industrial revolution.

The textile consumption also correlates strongly with the income levels of consumers. In underdeveloped and developing regions, as the income rises, the consumption tends to increase proportionally. However, in highly developed countries, once the income reaches a certain threshold, the rates generally stabilize.

Essential Needs for Education

Given the critical role textile goods play in our daily lives including the environmental sustainability, it is vital for everyone to gain a basic understanding of key aspects related to these materials. While some readers may already be familiar with the lifecycles of the textiles they consume from raw materials to manufacturing, distribution, consumption, and disposal as well as performance metrics and environmental impacts, many others may lack this knowledge.

With my educational and professional background, I am frequently engaged in casual discussions on a wide range of

textile-related topics, from environmental impacts to quality and maintenance, with those around me. From these conversations, I often encounter people who feel confident in their understanding of various textiles and their ability to make informed purchasing decisions. However, in my view, significant gaps are often found between their perceived knowledge and scientifically sound and sustainable practices. For example, an intentional choice of a supposedly more sustainable material can sometimes lead to unintended environmental harms. This chapter invites readers to delve into these issues and reassess whether their confidence aligns with the practices that truly benefit the environment.



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Section 1 Unfortunate Consumers

The **Section 1** explores several critical topics of misinformation and their sources, with a particular focus on the consumer perspectives. Initially, I will present broader viewpoints on global environmental management, highlighting how various environmental interest groups and activists often misplace blame on consumers and textile companies based on unscientific or invalid notions. Afterward, the discussion will shift to legitimate concerns about poor business practices within the textile industry and the consumer market, including those involving some of the largest and most reputable companies worldwide.

While I made an effort to provide detailed explanations backed by scientific information in relevant sections throughout the book, some of the brief descriptions and statements in the following discussion topics in the beginning may challenge many readers' existing beliefs. However, I encourage the readers to approach these topics with an open and inquisitive mindset, as there will be ample opportunities for deeper discussions and the scientific analyses of important environmental issues in various subjects ahead.

Consumers Targeted

In recent years, consumers, especially those in the Western world, have been the focus of many environmental interest groups, who criticize the overconsumption of apparel, particularly under the context of so-called "*fast fashion*". However, various research indicates that the global growth in textile consumption is primarily driven by "*the global*

population and income growth in emerging economies, with very little contribution from volume growth in Western markets"

https://emf.thirdlight.com/file/24/uiwtaHvud8YlG_uiSTauTUH74/A%20New%20Textiles%20Economy%3A%20Redesigning%20fashion%E2%80%99s%20future.pdf



Major Development in Consumption Trend

As mentioned earlier, the consumption rates of textile goods are closely related to basic human instincts. Throughout history, people have always sought to look good and express their identities through their appearances. As the public gained more economic power, it was a natural occurrence that the consumption rates increased. However, for the first time in human history, we are witnessing a plateau in the consumption rates in some affluent countries, as the satisfaction of personal desires begins to align with the quantity of goods people consume.

Furthermore, there have been encouraging signs that growing environmental awareness among the public is contributing to this significant shift in the consumption patterns. As this awareness grows, there has been a movement urging citizens in affluent countries to reduce their consumption of textile goods, while the global community accommodates a natural consumption growth in underdeveloped and developing nations.

Need for Effective Efforts

Although the trends in some developed countries show positive signs, significant challenges remain with textiles.

The cumulative effects of human living over thousands of years, particularly accelerated by the rapid industrialization and population growth, pose an imminent threat. This urgency is further underscored by the toxic impacts most textile goods create throughout their entire lifecycles - The concept of the toxicity mentioned here represents a perspective distinct from the prevailing framework that underpins much of today's global sustainability discourse and action, namely, the equation of carbon emissions with temperature rise and climate change, and constitutes one of the central arguments of this book.

Effectively addressing the environmental impacts of textiles to achieve an overall reduction is critical. Simply blaming global consumers at large for overconsumption is unlikely to be effective. Instead, more successful strategies will involve well-designed approaches that target consumers in different economic zones, combining education and methods that appeal to their psychological consumption behaviors and basic instincts. These tailored strategies can help create more positive environmental outcomes. This approach will be further explored in **Chapter V: Solutions**.

Companies Targeted

Many environmental advocacy groups often hold various companies in the global textile industry responsible for the environmental and sustainability challenges we face today. While it is undeniable that these companies create direct impacts on the environment through their operations, by fixating on corporate responsibilities, often with scientifically invalid bases, we risk overlooking deeper, instinct-driven consumer behaviors that contribute more

majorly to these issues. After all, consumers ultimately make purchasing decisions based on their own free will.

Virtue of Profit-Seeking Organizations

From the perspective of the profit-seeking organizations, it has long been viewed as a virtue to maximize their business outcomes in the form of profits. When there is a consumer demand, there is always a seller willing to create a system to meet those needs and turn it into profits. This is how the economics has functioned for thousands of years. Operational strategies of these companies might include a wide range of factors like point of sale, shipping method, marketing, technological development, and more, many of which are now under environmental scrutiny.

III-Conceived Targets

These targeting acts, however, often lack proper context. For example, blaming a "*fast fashion*" company for using air cargo, which generates higher carbon emissions than sea freight, must be contextualized against the environmental impacts of "*slow fashion*" companies. The later form of business often operates large storefronts that consume substantial energy for lighting, heating, and more, along with the carbon emissions generated by their employees commuting to and from these locations and the resources needed to maintain such corporate systems. In reality, we lack sufficient data to fairly compare the overall environmental impacts of these different operational models. This example illustrates how many environmental advocacy groups frequently present information selectively, focusing on the points that support their arguments without considering the full picture.

Hidden Sides

In the public eye, some companies may seem more ethical by adopting environmentally or socially responsible business practices in certain aspects of their operations. However, throughout my career, I have often seen that even some of the most reputable global companies prioritize promoting their positive images, while failing to live up to their claims, promises and often engaging in what can only be described as greenwashing. In various discussion subjects of this chapter, I will present several concrete examples of such corporate misconducts.

Fair and Comprehensive Judgement

Conversely, conventional judgments on various environmental and sustainability issues often rely more on consumers' general perception of a certain brand and/or the retail values of their goods (expensive vs. inexpensive), business practices, etc., rather than on true environmental impacts based on accurate science. In more recent years, certain trade conflicts have led to influencing public judgments on the goods originating from some countries under the context of the environment and sustainability.

However, it is important to emphasize that the true environmental impacts are rarely confined to singular or overly simplistic aspects and notions. Our evaluations must be comprehensive and grounded in accurate scientific analysis. Without a balanced and fair approach to the judgments, targeting specific entities or practices under the banner of sustainability can be misguided and may not lead to desired improvements in environmental outcomes.

Overproduction Issues and Safety Hazards

For example, recent overproduction issues in China have become a major topic in the business world and are often connected to various environmental concerns. Furthermore, there have been numerous reports of safety hazards associated with products purchased from online platforms that offer extremely low prices. These products, which include toys, personal accessories, plastics, textiles, cosmetics and more, often contain harmful chemicals that exceed the safety limits set by different countries.

This issue highlights a significant lack of quality control and moral ethics among the manufacturers, exporters, importers, and platform operators involved in such incidents. These safety hazards not only pose risks to consumers' health but also have detrimental environmental impacts.

Regardless of the environmental concerns, however, such business practices should not occur in the first place as they are illegal and intolerable. Unfortunately, these illegal practices are not limited to low-cost value chains as they have also been found, though presumably less frequently, in products sold by even reputable companies with higher price tags on their goods. These incidents may happen with or without the knowledge of some or all of the involved parties, primarily driven by profit motives and/or lack of efforts.

Need for Accurate Assessment

In my view, trade conflicts, unfair business practices, extremely low prices and safety hazards should remain focused on economic, legal, and political debates, rather

than being conflated with environmental and sustainability issues. For example, if a country determines that certain goods sold on a particular platform are likely to contain harmful chemicals that threaten the health and safety of its citizens, necessary actions should be taken to prevent such risks, independent of other considerations. While it is true that environmental topics tend to capture public attention more effectively and elicit stronger reactions these days, this approach can obscure the core issues that need to be addressed for the environmental sustainability.

1.1 Material Choices and Care Methods

One of the biggest areas of misunderstanding in textiles, driven by widespread misinformation, involves the public perceptions of different material types and their environmental impacts. For example, many people believe that natural fibers like cotton, wool, and silk are more sustainable than synthetic fibers like polyester, Nylon, and polypropylene. This perception is rooted in the following three common perspectives:

- A. **Origin Factor:** *“Synthetic fibers are produced from petroleum sources, creating carbon footprints, whereas natural fibers are sourced from the nature”.*
- B. **Decomposition Factor:** *“Synthetic fibers persist in the environment much longer (some argue indefinitely), while natural fibers decompose quickly”.*
- C. **Microplastic Factor:** *“Synthetic fibers shed microplastics, contaminating our food sources, leading to bioaccumulation in humans and animals, and causing health issues”.*

These seemingly straightforward arguments resonate with the public and have shaped the global environmental mindsets and associated efforts. While some of these points are accurate in isolation, they often lack numerous critical contexts needed for a more comprehensive assessment of the true environmental impacts between the two fiber categories. Below are some of the examples:

A.1 Origin Factor: *“Synthetic fibers are produced from petroleum sources, creating carbon footprints, whereas natural fibers are sourced from nature.”* : This argument critically overlooks the significant energy consumption involved in farming natural fibers and, in some cases, the amount of the carbon footprint created in natural fibers can be multiple times higher than that of synthetic fibers - Detailed discussions on carbon emission comparisons amongst different fiber types will be presented with a set of scientific data from a research paper in **Chapter III: 4.4 Carbon Footprint Index (CEI) Analysis.**

B.1 Decomposition Factor: *“Synthetic fibers persist in the environment much longer (some argue indefinitely), while natural fibers decompose quickly.”*: The notion that natural fibers decompose quickly considers the breakdown of the natural elements like the cellulose of cotton and the protein of wool. However, it fails to account for the residual chemicals in textile waste. The quantity of harmful chemicals contained in the fibers, their toxicity and persistence in nature are mostly overlooked, yet critical factors for evaluating the overall environmental impacts. This topic is one of the central themes of this book and will be explored in depth throughout the book,

including the dedicated **4.2 Decomposition Myth** discussion later in this chapter.

C.1 Microplastic Factor: *“Synthetic fibers shed microplastics, contaminating our food sources, leading to bioaccumulation in humans and animals, and causing health issues.”*: While synthetic fibers do release microplastics throughout their lifecycles, this argument critically overlooks the issues from natural fiber dust. Natural fibers, like synthetics, shed dust during their lifecycles. The quantity of the natural fiber dust and the amount of harmful chemicals contained in it compared with the synthetic microplastics may surprise many readers - A detailed analysis on this subject will be provided in **3.2 Dust of Natural Fibers** later in this chapter.

Furthermore, many other critical factors are missing from the global perception of the environmental impacts between natural and synthetic fibers. To keep the discussions more relevant for each subject throughout the book, I will present these factors in the appropriate sections and subsections. In this section, I focus on the topics more closely related to the consumption of textile goods, starting with the discussion on the care methods:

1.1.1 Care Method Overview

Care instructions on the woven labels attached to most textile goods vary based on several factors, with the most significant being the recommended cleaning method, either home washing or drycleaning, often determined by material types, whether natural or synthetic. These methods are intended to help consumers preserve the appearances,

shapes and fits of their textile goods in their original conditions for as long as possible. Additionally, there are other factors related to the care and maintenance that are not typically incorporated in the care instructions. This discussion will explore these important considerations, which not only affect the longevity of textile goods during use but also create significant environmental impacts.

Micro-Organismic Activities and Frequencies of Care

Material types can heavily influence the care frequencies during consumption period. For instance, cotton shirts and rags tend to develop odors more easily than their polyester counterparts, a phenomenon familiar to many. This observation aligns with the scientific understanding of the natural fibers and their chemical compositions, such as the presence of hydroxyl groups in the plant-based fibers (cotton, linen, etc.) and proteins in the animal-based fibers (wool, silk, etc.), which inherently offer high chemical reactivities with odor-causing microorganisms - A more detailed exploration on this topic is provided in **Chapter II: Basics**. It is therefore a common practice to wash or dry-clean textiles made with natural materials more frequently than their synthetic counterparts.

Color Fading and Physical Deformation

The natural fiber textiles are often recommended for drycleaning due to their higher susceptibility to color fading and physical deformations, such as shrinkage and stretching, when washed with water at home. These issues stem from the higher chemical reactivity of the natural fibers, combined with their lower inherent strength. This is particularly true for the animal-based materials like wool,

silk, leather and down, which frequently come with drycleaning instructions - The toxic nature of drycleaning chemicals and their environmental impact will be discussed in detail later in this section.

In the following discussions, we will explore the principles behind different care methods for various fiber types and their corresponding environmental effects.

1.1.2 Natural Materials and Care Methods

Numerous natural materials are used in textiles, primarily categorized as either animal- or plant-based. In this subsection, we will explore three of the most commonly used natural materials: Cotton, Wool and Down.

A. Cotton

Cotton's affinity for water and the heightened chemical reactivity of its hydroxyl groups present significantly different characteristics compared with synthetic fibers in maintaining the appearance, shapes and fits of the textiles made with the material. For instance, most readers may have noticed that their cotton shirts tend to lose color more readily than their synthetic counterparts after multiple home washes. In knitted products such as sweaters, T-shirts, etc., it is also common to observe necklines becoming loose especially if hung-dried or a considerable shrinkage occurring when the temperature setting of washing machine and dryer was too high. All these phenomena can be explained by the chemical compositions and strength factor of cotton in comparison with its synthetic counterparts.

While most textiles containing cotton can be maintained with careful home wash, some individuals opt to dry-clean their valuable cotton garments to prevent color fading and deformation mentioned earlier. Since drycleaning does not involve water, it significantly reduces the risks of such occurrences.

B. Wool

Wool fibers offer a unique surface structure with the scales in the linear alignments of the fiber's length direction, as shown in **Fig.3**.

Fig.3 Scales on Surface of Wool Fibers (coarse and fine)



<https://www.silhouettetailoringstudio.com/shrinkage/>



Similar to fish scales, the scales on wool protect the internal protein structure beneath. Unlike the protein of fish, however, wool's protein molecules absorb a large amount of water and swell when submerged in water. As a result, the outer edges of the scales extend and protrude more outward. During a home wash with water, the physical agitation inside the washer causes the scales, aligned in opposite directions, to interlock with each other as they act

like tiny hooks. This process, known as “felting”, leads to shrinkage of woolen items in home laundry. Once the felting occurs, it cannot be undone, effectively ruining the item. This is why the care instructions of woolen goods typically recommend drycleaning.

In some cases, woolen items are labeled as washable in water. However, this requires a process that dissolves the tip parts of the scales in strong chemical solutions. While it can eliminate the need for drycleaning, it introduces potentially much more harmful chemicals during manufacturing process.

C. Down

The down-feather material, commonly known as “down” and originated from various types of birds, presents a higher level of complexity than other textile materials, primarily because it is typically used in more intricate constructions that involve multiple components, such as enveloping outer fabrics and insulating layers which contain the material. These complex structures can be easily understood when comparing between a winter jacket with down material located deep inside the structure, thus invisible, and a single-layered shirt or pants.

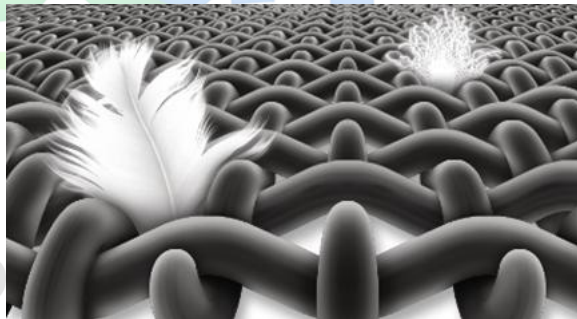
Instructional Variations: Textile goods that utilize down materials as insulator fall into two categories based on their care instructions - those that are instructed to be drycleaned and others recommended for home washing. This discrepancy often leads to questions and confusion, particularly given that the primary goal in both scenarios is to clean and maintain the performance of the valuable down material.

Dry Cleaning Recommendations

Following factors are considered for the drycleaning recommendations:

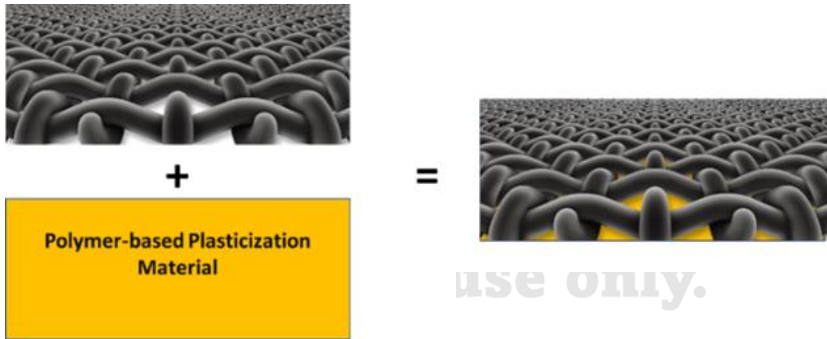
Drying Time and Fabric Treatment: Washing textile goods containing down materials at home can require multiple drying cycles unlike most other items which will dry in just one cycle. While the complex structure mentioned earlier is a contributing factor, this extended drying time can be primarily due to the plasticized fabrics used to prevent down leakage - The structure of woven fabrics enveloping down material creates small holes at the intersections of the crossing yarns, known as warp and weft. Although these holes are small in sizes, they are larger than the fine hairs of down feathers and can become the channels of escape, as illustrated in **Fig.4**.

Fig.4 Down Leakage



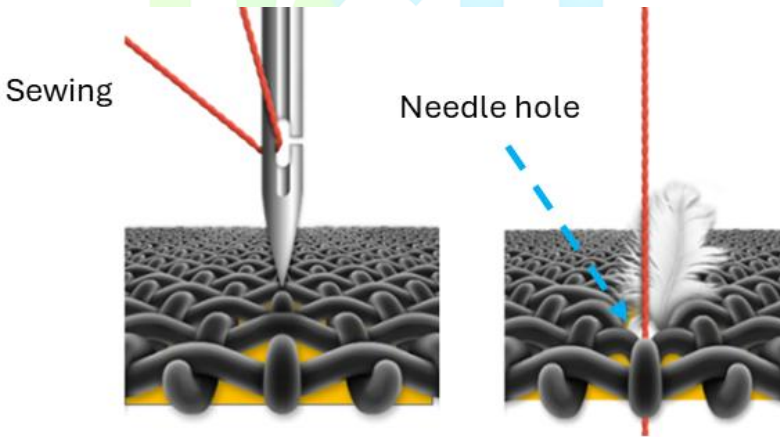
To mitigate this, the fabrics used to create the containment of down materials, commonly known as “down bags”, are often plasticized to close off the holes as shown in **Fig.5**.

Fig.5 Plasticization and Down Leakage Prevention



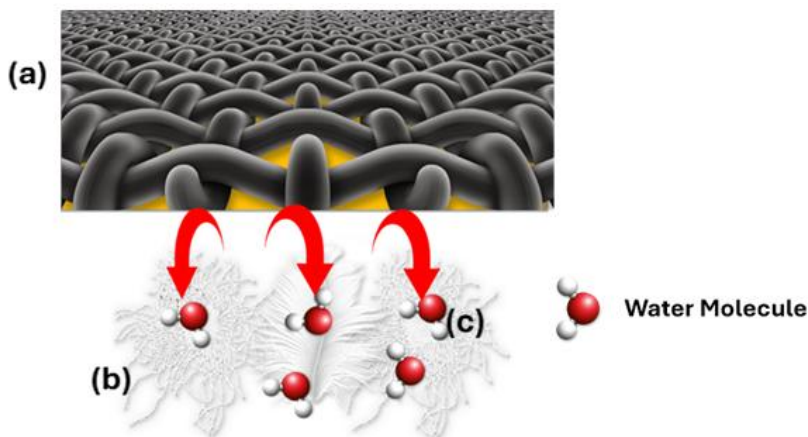
Despite this effort, consumers still experience down leakage often through the needle holes as depicted in **Fig.6**.

Fig.6 Down Leakage through Needle Holes



This plasticization covers much of the fabric, significantly slowing moisture evaporation rate during drying process by blocking the escape of water molecules, illustrated in **Fig.7**.

Fig.7 Water Molecules Bouncing Back



(a) Plasticized fabric surface (b) Down material
(c) Water molecules resisting evaporation

Consequently, drying a down-filled product at home can be a half-day chore, requiring multiple drying cycles, often with tennis balls to agitate the down material, ensuring that it dries thoroughly from the inside out. Inadequate drying can leave the material damp deep inside the structure, creating an ideal environment for microbial growth, fueled by the protein in the down feather and the nutrients from our own sweat, such as urea, ammonia, and fatty acids. Although many people may not recognize, “down” items such as garments, blankets, sleeping bags, etc., can be the sources of various health issues if improperly maintained. Acknowledging these challenges, many brands recommend drycleaning.

Appearance: Drycleaning instructions can be motivated by other reasons. For instance, if a down jacket contains metallic components such as badges, zippers, buttons, and snaps, washing it with water and detergents can lead to significant discoloration and unwanted aesthetic changes.

Drycleaning helps mitigate this issue. In such cases, the drycleaning instruction is unrelated to the care or performance of the down material. However, some notable brands avoid using metallic components altogether. Instead, they use embroidered textile-based badges, along with plastic components for zippers, buttons, snaps, etc. This allows them to safely recommend home washing to their consumers, reducing the significant environmental impact of toxic drycleaning chemicals while also offering considerable cost savings for maintenance.

Warmth Performance affected by Cleaning Methods:

Down's delicate structure, made up of numerous fine hairs, is more vulnerable to damage from the physical agitation of home washing mechanisms. When down feathers absorb water, they become considerably heavier, increasing physical forces acting on them during washing, which raises the risk of damage and ultimately reduces their warmth performance. Drycleaning, which does not involve water, is gentler on down. However, the effects of infrequent home washing, such as once or twice per winter season, often go unnoticed by untrained consumers, as a sufficient amount of down is typically inserted during manufacturing to maintain a certain level of warmth over a considerable period of use.

Effectiveness of Cleaning Methods: In the perspective of the effectiveness of the two different cleaning methods, many pieces of conflicting information are found on the internet, making different claims, some insisting that down materials need to be drycleaned while others favor home wash. Each argument proposes a reasoning or two. Rarely,

however, these discussions include the nature of contamination in relation with the care methods.

In essence, home wash is effective if a contaminant is soluble in water. If the contaminant is not water-soluble, certain drycleaning chemicals may be more effective at removing it.

Nature of Contaminants: A large proportion of contaminants for down materials are water-soluble, making home washing with suitable detergents sufficient for their removal. However, exceptions exist. For example, cooking oil splashes onto clothing. Depending on the type of oil and the nature of the contaminated fabric, these stains can be stubborn and resistant to home washing. Dry cleaners may use specific chemicals that are effective for such oil-based contaminants.

That said, in everyday use, it is unlikely that down materials will be exposed to water-insoluble contaminants, as they are typically protected by layers of enveloping fabrics. Even in an unlikely scenario where someone cooks while wearing a winter jacket and oil splashes onto it, the chances of the oil penetrating through the outer layers and contaminating the down are minimal. The most common contaminants for the down materials in such protective structures are the compounds contained in our sweat, as they penetrate through the fabrics and reach the material. These substances are water-soluble and can be effectively removed through home washing.

- * Special mention – Leather, while it is one of the most commonly used natural materials, is not covered more extensively in this discussion largely because they are invariably instructed for drycleaning. It is a common

knowledge that home-washing leather goods can create significant discoloration. This phenomenon relates to the inherent chemical reactivity of both the leather itself and the chemicals used to colorize it, which will be further discussed throughout the book.

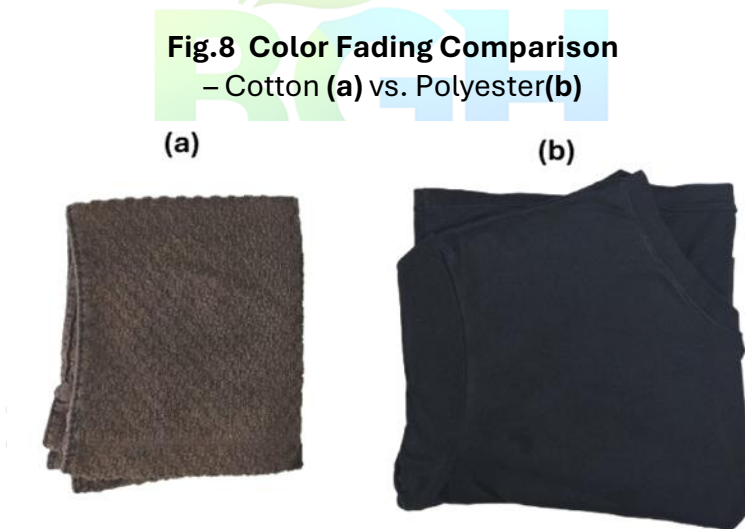
Chemical Toxicity of Drycleaning Chemicals

A surprisingly large number of people are unaware of the toxicity associated with drycleaning chemicals. Perchloroethylene (PERC), one of the most commonly used solvents in drycleaning around the world, is a reproductive toxicant, neurotoxicant, potential human carcinogen, and a persistent environmental pollutant. I vividly recall one of my textile engineering professors warning our class, *"If you live in a building where there is a drycleaning shop, move before it kills you!"*. His opinion, while it may be debatable given many variables from one building to another with different exhaust systems, etc., was to alert the students of the nature of most solvent-based cleaning chemicals as odorless gas that spreads quickly and widely, making it particularly hazardous.

Nature of Solvents: Although efforts have been made to develop less toxic solvents for drycleaning, ongoing debates persist about the sustainability and safety of these alternatives. Drycleaning solvents, by their very nature, are chemically potent enough to alter the states of contaminants, detaching them from contacting textile materials by chemical interactions without water. This potency inherently makes them hazardous to both humans and the environment.

1.1.3 Synthetic Materials and Care Methods

Different synthetic materials exhibit varying characteristics in terms of their environmental impacts. While detailed information on these overall environmental effects will be thoroughly examined in **Chapter II: Basics**, one significant advantage of most synthetic materials is that they typically do not require drycleaning. This distinction, compared to their natural counterparts, stems from a significantly lower chemical reactivity, which gives them a strong resistance to color fading. Furthermore, they offer much higher strength, enabling them to withstand physical forces encountered in home laundering with water. **Fig.8** below illustrates one such example.



The cotton towel shown in (a) is one item from a set of twelve hand towels that I purchased approximately **two years ago**, while panel (b) shows a polyester T-shirt that has been in use for nearly **twenty years**. It is reasonable to estimate that both items were laundered at roughly the

same frequency, approximately once every one to two weeks. Notably, at the time of purchase, both products were of a similarly deep black tone. A comparison of their current coloration reveals a striking difference despite the substantial disparity in their periods of use. The pronounced color fading observed in the cotton product can be attributed to the detachment of dyes and other chemical substances during repeated laundering, which are subsequently washed away with the water - Understanding this difference also implicates a wide range of scientific principles including fiber strength and chemical contents in each fiber types. This will be explored in various sections of this book.

After reviewing several key factors associated with different care methods and their environmental impacts across various textile material types, following discussions will focus on specific technologies used in consumable textile products and their potential environmental impacts:

1.2 Water Resistance & Breathable Technology

Many outdoor enthusiasts may recognize this technology, as it is widely used in outdoor apparel and is also commonly found in workwear, tactical gear, and other applications. This subsection explores how the technology functions and its potential environmental impacts.

Definition: Before diving deeper into this topic, it is essential to clarify a common misunderstanding regarding the terminology. ‘Waterproof’ is a loosely applied term that

relates to the performance of the fabrics with this technology. In a more precise scientific description, it refers to a resistance up to a certain level of water pressure. Therefore, the term "water resistance" is more accurate than 'Waterproof', as beyond a certain pressure threshold, the fabric will leak water. Applying a similar logic, many other fabrics can be labeled as 'Waterproof'. For example, tightly woven synthetic fabrics without this technology can resist water up to a certain pressure. The key difference lies in how much water pressure a fabric can withstand before allowing water to seep through. Despite the discrepancy, the term 'Waterproof' for the fabrics employing this technology has been used in the industry for decades and is widely accepted by both sellers and consumers.

'Breathability' refers to the fabric's ability to allow moisture-laden air from the body to pass through (while providing a high level of water resistance in the context of this technology platform), providing dry comfort to wearers.

Confusion with Water Repellency: Additionally, some consumers confuse water resistance with "water repellency". Companies promoting 'Waterproof' feature often use images showing water droplets beading on the surface of a fabric, the phenomenon unrelated with the 'Waterproof' claim. While both water repellency and *waterproof* features give the appearance of resisting water, they differ significantly in the chemicals used and the areas where the chemicals are applied – The 'water repellency' will be explained after the ongoing discussion.

Technology Principles: The 'water resistant and breathable' technology involves applying a polymeric membrane, typically made of the expanded-

polytetrafluoroethylene (ePTFE), to the fabric's backside (the side opposite to the weather-facing surface). This membrane contains countless tiny pores that are small enough to prevent water droplets, such as those from rain, held together due to the surface tension of the comprising water molecules, from passing through. However, the pores are large enough to allow moisture vapor from sweat, which consists of smaller water molecules, to escape, achieving both the water resistance and breathability.

That said, under enough pressure and depending on the size of the pores, the water on the fabric surface can eventually overcome its own surface tension and penetrate through the membrane, allowing it to reach the wearer's skin. This phenomenon is better understood by reviewing a typical test method used to evaluate the performance metrics.

Test Method: A piece of fabric is placed flat within a cylinder that separates the top and bottom compartments with the fabric dividing the two. Water is then poured into the top compartment, while the bottom compartment remains empty to allow the observation of water droplets passing through the fabric. Test equipment typically includes a pressurization system that simulates water pressure at specific levels, such as 5,000mm (or 5 meters / approx., 17ft), 10,000mm (or 10 meters / approx., 33ft), or higher.

As the water pressure increases, the fabric eventually allows the water to seep through, and droplets begin to fall into the bottom compartment. The tester records the water pressure at the point when a certain number of droplets are observed, which determines the fabric's water resistance level. For example, a 10,000mm rating indicates the fabric

can withstand the equivalent of 10,000mm of water pressure.

Effectiveness in Extreme Conditions: This technology has gained a widespread popularity globally, with many people choosing garments and outdoor gear that feature it. For avid outdoor enthusiasts or professional explorers facing extreme wilderness conditions, high levels of water resistance can be crucial for survival. For instance, a 10,000mm performance level can be lifesaving in places like the Himalayas or the South Pole.

Phenomenon in South Korea: In South Korea, a particular brand known for this technology has become so popular that it is now applied in a wide range of apparel items, including large quantities of everyday clothing, regardless of wearers' activity levels or weather conditions they typically encounter. Students, grocery shoppers, casual walkers, and light hikers, including my own friends and family, frequently wear garments with this technology. During a business trip to South Korea, I visited a department store with my family and my sister was looking for a new jacket with this water-resistant feature, more specifically, one with the famous brand's logo. When I asked if she understood the technology and why she needed it, her response was simple: *"It is widely popular and everyone around me wears it. There must be a good reason for it, and it should apply to me."* Given that she is a working mom whose outdoor activities rarely extend beyond grocery shopping, it was clear to me that her desire for the feature defeated the purpose of this technology.

Phenomenon in Canada: More recently, I had a similar experience with a friend in Canada, an avid runner who regularly participates in half and full marathons. One day, he joined in a light hiking event I organized on a sunny day, and was wearing a pair of running shoes that incorporated the same brand as the one my sister was looking for. The brand's name, prominently displayed on the sides of the shoes, caught my attention. When I asked about his understanding of the technology, his response was remarkably similar to my sister's.

I have noticed this trend in many other countries I travel to, especially in affluent regions, where this expensive technology is often used in everyday clothing.

Practicality of Performance: One internet blog I came across claimed that a 10,000mm water resistance rating is equivalent to standing under a heavy waterfall for 30 minutes. Although I could not verify the accuracy of this claim, given such vague term as “heavy” waterfall, it is easy to understand that 10,000mm (10 meters or 33ft) is an extremely high level of water pressure for most people in their everyday activities. Personally, as a middle-aged man who considers his lifestyle to be more or less average in the general population, I have never encountered a situation where such a high level of water-resistant performance was necessary, even including during my younger years when I frequently enjoyed outdoor activities like intensive hiking in mountains and camping in Korea.

Sweat Challenges: Some of my friends and family members have mentioned that when they wear their jackets with this technology, they often feel a quick buildup of sweat inside, even during a short hike. This contradicts the

high breathability perception consumers gain on these products as a result of associated promotional information. Scientifically, this observation is accurate. When the level of breathability cannot keep up with the amount of sweat produced by the body, the vapor accumulates inside the garment. How quickly this happens depends on various factors like wearer's metabolism, activity level, health, and weather conditions. While the membrane contains numerous tiny pores, the rest of the fabric is blocked for the vapor's escape. If the sweat production exceeds the membrane's capacity to release it, the accumulation occurs.

Lack of Understanding: Choosing between increased water resistance with reduced breathability offered by this technology and maximizing breathability without this technology is a personal choice. However, in my opinion, preparing for extreme weather events combined with prolonged outdoor activity, that may never occur for many consumers like my sister, might not be the most logical approach. While this technology can be valuable in situations where highly pressurized water is a concern, such as in footwear, where our feet bear the entire body weight on a small surface area, or in the areas like elbows, knees, and buttocks of ski wear or workwear for specialized job functions involving kneeling, elbowing on wet surfaces, etc., its practicality for everyday clothing is what concerned me during my conversation with my sister at the department store. Unfortunately, many people are unaware of the trade-offs including the environmental impacts of the technology and are often influenced by social trends rather than their actual needs.

Environmental Impacts: Applying a polymeric membrane to the backside of fabrics, a process often known as ‘plasticizing’, makes these fabrics unrecyclable with currently available recycling technologies. The fabric and membrane create a tight bond, which cannot be separated, significantly compromising the recyclability of these materials. Since successful recycling relies on the purity of the materials in question, this technology could result in large quantities of fabrics becoming unrecyclable, even though they could otherwise be efficiently processed using the **Clean Recycling Initiative™** technology platform. More detailed discussions on this platform will be provided in **Chapter V: Solutions.**

Additionally, materials like ePTFE used in this technology are incredibly durable. They are dense polymers and are some of the heaviest materials on Earth by volume. While I will address the widespread misunderstanding about plastic degradation later in this chapter, it is important to note that some plastics last longer in nature than others and their densities play a key role. As a result, textiles incorporating this technology may persist in the environment for much longer than the equivalents without it.

Debates: I have witnessed many debates regarding the environmental impacts of this technology. One stakeholder, in particular, made an argument that the technology extends the lifespan of the fabrics by making them stronger, assisted by the membrane, thus positively impacting the environment.

In analyzing these claims and debates, it is important to note that the fabric strength is rarely a reason contributing

to the disposal of textile goods these days, particularly in affluent Western countries. This is evident when examining textile items deposited in community centers, recycling bins or donation depots as they are often in wearable conditions, sometimes needing only minor repairs like fixing loose buttons or broken seamlines, more of which will be discussed in the **Unfortunate Disposal** later in this chapter.

Consumer Awareness: When purchasing everyday textiles, my decisions revolve around the likelihood of encountering certain weather events and my plans for dealing with them. Based on these considerations, my personal shopping habits prioritize efficient sweat removal and the environmental impacts of the goods I purchase. As mentioned earlier, I have never encountered a situation where 10,000mm of water resistance was essential. In the event of sudden, heavy rain, I seek a shelter. For a forecasted weather event, I carry an umbrella. This strategy has always worked for me, so my wardrobe does not reflect improbable scenarios that go beyond my life experiences and action plans. If, one day, I were to embark on an extreme expedition, I would carefully strategize my textile choices, and the high-water resistance and breathable technology might become a priority.

1.3 Water-Repellency

Water-repellency in textiles relates to the surface tension of fibers, fabric structures and their relationship with that of contacting water molecules. This phenomenon can be easily observed in various objects such as water beading on lotus leaves or certain fabrics. The surface structure of, for example, lotus leaves creates a level of surface tension

insufficiently strong to break the internal tension of water molecules, causing water droplets to maintain their shapes.

Cotton and Wool: Cotton fibers and fabrics have a high affinity for water due to the presence of the hydroxyl groups in cellulose. This causes water to penetrate and spread quickly through the materials. Wool, on the other hand, behaves differently because of its unique surface structure. While the protein in the wool also has a strong affinity for water, it is protected by the scales, as shown in **Fig.3**, on the fiber surface that provide a certain level of water repellency and resistance. However, once water penetrates through the gaps of the scales and wets the protein, it quickly spreads.

With their inherent affinity for water, fabrics made with cotton and wool may require chemical treatments to provide a desired level of water repellency, if used in textiles designed for direct contact with rain or snow.

Synthetic Materials: In contrast, tightly woven synthetic fabrics, even without chemical water-repellent treatments, can behave like lotus leaves to a certain degree, causing water droplets to bead up on the surface. This natural water repellency is closely related to the hydrophobic nature and smooth surface structure of most synthetic fibers.

Chemical Treatment: Per- and polyfluoroalkyl substances (PFAS) are applied to the weather-facing side of fabrics to create a high(er) degree of water repellency, preventing rain from wetting the surface. This treatment is commonly used on textiles made with natural fibers such as a woolen (brushed) winter coat, cotton jackets, etc. Another example is various outdoor jackets made with synthetic fibers.

Although synthetic fibers naturally repel water, the chemical treatment boosts their performance to a level more suitable for avid outdoor activities.

The PFAS are organic compounds that come in different variations, such as C8, C6, and C4, with the numbers referring to the number of carbon atoms. Traditionally, the industry used the compounds with higher carbon numbers, like C8, also known as perfluorooctanoic acid (PFOA), because they offer superior water repellency and durability on textile materials compared to the compounds with fewer carbons.

Toxicity Recognition and Regulatory Works: In the late 1990s, several lawsuits were filed against DuPont by residents living near its Washington Works plant in Parkersburg, West Virginia, due to a PFOA contamination originated from the plant in the local drinking water. Subsequent research identified the PFOA as carcinogen, with a particular link to kidney and testicular cancers.

In 2006, the U.S. Environmental Protection Agency (EPA) and eight major companies that used these chemicals in various industrial applications agreed to the 2010/2015 PFOA Stewardship Program. The program's goal was to reduce PFOA emissions by 95% by 2010 and to work toward eliminating them by 2015. This initiative was successfully implemented, and by the mid-2010s, these companies had phased out the use of the PFOA in manufacturing consumer products and other intermediary materials such as fibers, yarns and fabrics.

In June 2020, the EPA issued the Significant New Use Rule (SNUR), requiring US manufacturers and importers to notify the agency before beginning or resuming the use of these

chemicals for any significant new purposes. Following the EPA's actions, the European Union also took major steps to regulate these chemicals. In 2019, many countries participating in the Stockholm Convention agreed to add the PFOA to the list of the Persistent Organic Pollutants (POPs).

Continued Use and Environmental Impacts: Despite these regulations, various PFAS chemicals are still used in many countries where environmental regulations are less strict. Additionally, significant loopholes exist in the regulatory actions of many countries. For instance, the EPA's action against the PFOA (C8) led to the use of other PFAS with shorter carbon chains, such as C6 (perfluorohexanoic acid, PFHxA) and C4 (perfluorobutane sulfonate, PFBS). However, concerns on the health impacts and the persistence of these chemicals in nature remain, as they belong to the same family of chemical groups.

Several studies have revealed the potential health impacts of the chemicals with fewer carbon atoms, such as thyroid hormone disruption, developmental toxicity, liver toxicity, and effects on the immune system at high doses. Even without these studies, scientific reasoning suggests that, while the impacts of C6 and C4 may be less severe than those of C8, they still exist and create proportionally harmful impacts.

These chemicals contaminate water either from manufacturing processes or laundering the textiles containing the chemicals, then spread globally through water streams. Moreover, the PFOAs applied to a large number of consumer and industrial products before the implementation of the environmental regulations still

persist in the environment due to their extreme durability. Combined with the ongoing use of these chemicals in many regions of the world, they will continue to impact the environment and human health for generations.

Consumer Awareness: I have often been caught in the rain before finding a shelter in my everyday clothing, and the feature could have provided some relief by keeping my body dry in such situations. With this, I recognize that the water repellency might be more useful for people like me in their city life than the water resistant and breathable technology. However, I still weigh my decisions carefully: These chemicals gradually wash out with each laundry cycle, eventually losing their effectiveness. The preference for C8 compounds over C6 and C4 largely stems from their superior durability on fibers and fabrics. Nevertheless, no matter how durable, this performance feature usually has a shorter lifespan than the textile item itself. In some cases, the repellency lasts only few washes, depending on wash conditions.

Justification: Given the limited duration of its effectiveness, as well as the associated costs and environmental impacts, it is hard for me to justify purchasing items employing this chemically treated feature. For these reasons, I actively seek out the products without any water-repellent treatment.

As the section title suggests, consumers are often "unfortunate" because they are frequently targeted by various stakeholders who promote a range of technical claims, making certain features seem relevant when, in

reality, they may not be as beneficial as advertised. Additionally, consumers' choices are often limited by the decisions made by these stakeholders. For example, when shopping for a jacket, it can be difficult to find options that don't include unnecessary chemical treatments. Even if a shopper prefers to avoid these features, they may still end up purchasing a jacket with extensive chemical treatments, as these features are routinely applied to many textile products.

Unfortunately, consumers' limited understanding, coupled with a lack of interest in learning about the science behind these technologies, enables many organizations to prioritize performance claims over health and environmental concerns. This highlights the importance of consumer education. It is crucial for consumers to be informed about the practicality of certain performance features and to ask relevant questions about their environmental impacts. Given the importance of this issue, I explore it further in **Chapter V: Solutions**, particularly in the section titled **Education, Education, and Education**.

1.4 Artificially Conceptual Value (ACV)

An Artificially Conceptual Value (ACV) is a performance feature, that operates similarly to a placebo effect in medicine, but it can lead to unfavorable outcomes for the environmental sustainability.

Definition and Corporate Behaviors: To further define Artificially Conceptual Value (ACV), it refers to a technical feature that lacks scientific evidence to prove its effectiveness in real-life situations but makes people believe it works. Companies engaged in ACVs often

introduce related scientific concepts to explain how their technologies theoretically function. Unfortunately, these explanations are frequently designed to quickly convince consumers who lack in-depth knowledge about the related science. This approach is highly effective because it seems logical and plausible on the surface.

Consequences: In the medical world, relying solely on placebo effect can worsen an illness over time, leading to serious consequences. For example, it is well known that Steve Jobs could have potentially saved his life by turning to modern medical treatments rather than relying on naturopathy for too long. The same logic applies to the environmental impacts of various ACVs. A scientifically accurate evaluation can help remove potentially harmful and unnecessary chemical treatments in the textile goods we consume for the good of environment.

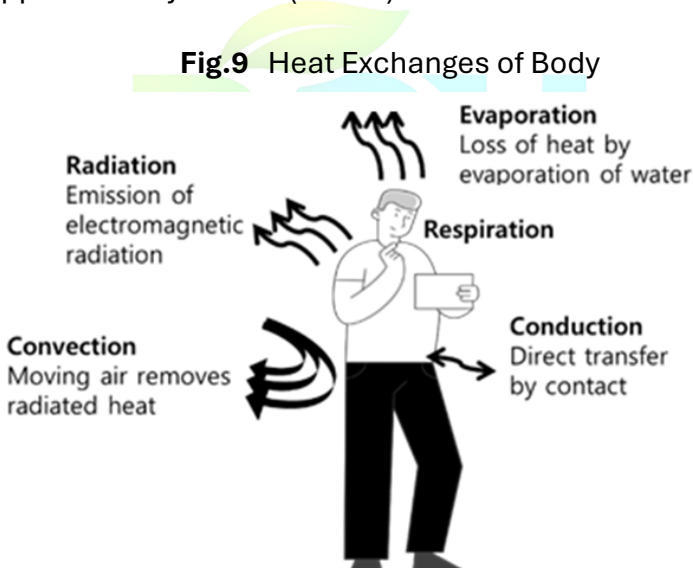
Unfortunately, ACVs are widespread in the textile industry. I'm often astonished to see large, reputable, and respected companies engaging in such practices. In this subsection, we will review several examples of ACVs in textiles to help readers better understand this concept.

1.4.1 Heat Reflective Technology

A thin layer of a metallic silver pattern is applied to the inner lining fabric of winter jackets with the claim that it reflects the body heat back toward the wearer, conserving the heat and keeping him/her warmer than the person would be without this technology. This concept sounds logical and plausible to many, and as a result, the technology has

become commercially successful in cold-climate regions where effective thermal technologies are essential for winter garments.

Heat Exchange: To understand the scientific foundation of this technology, it is essential to understand how the heat exchanges occur in the human body through five different mechanisms: conduction, convection, radiation, respiration, and evaporation, as illustrated in **Fig.9**. These processes help prevent overheating by dissipating the heat generated from the food we consume, which raises body temperature, and helps maintain a constant temperature of approximately 36.5°C (97.7°F).



The degree of the heat exchanges depends on several factors, including weather and individual body conditions, where some of these factors can be controlled, while others cannot. Without delving too deeply into the science of thermodynamics, simplified concepts are as follows:

Respiration, Convection, Conduction and Evaporation:

Given a specific set of weather conditions and personal variables, our body heat escapes through respiration as we breathe. Wind removes the heat through convection, as though standing in front of a fan. A conductive heat loss happens when cold air molecules come into contact with the molecules warmed by our body heat, transferring the heat from the higher energy level of being warm to the lower energy level of the cold surroundings. An evaporative heat loss occurs through the sweat glands on the skin to release the heat through moisture, which evaporates and cools the body. Increased physical activity, such as running, generates more heat and moisture, leading to visible sweat and greater heat loss through this mechanism of evaporation.

Radiation: A radiative heat exchange, the focus of this heat-reflective technology, involves the transfer of the body heat through radiation. This type of heat exchange can be easily experienced by standing near a radiator in a room during winter; you can feel the warmth radiating upwards when the radiator is on. Similarly, someone engaged in intense physical activity, like vigorous running, can feel this radiative effect by placing her/his hand near the forehead. When the body needs to dissipate excess heat quickly, it radiates the heat through the skin, working in tandem with other mechanisms like heavy breathing and perspiration.

Mechanisms of Controllable Heat Exchanges: Of the five heat exchange mechanisms, three - radiation, respiration and evaporation - are mostly under the control of our body, while conduction and convection depend more heavily on weather conditions. These three 'controllable' mechanisms

are most active during a high physical activity in warm and humid conditions as the body needs to remove the heat more efficiently in those conditions.

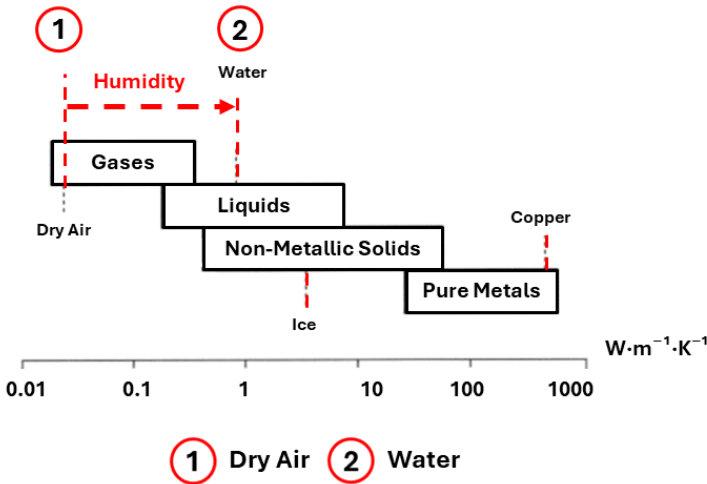
Conversely, in a cold weather with a low activity level, the body attempts to conserve the heat by minimizing these three exchanges. In such cases, the respiration slows, the body's sweat pores shrink and close, and the skin works to prevent the heat from escaping, reducing the radiative heat loss to nearly negligible levels.

Hypothesis and Scientific Analysis: If the heat reflective technology were truly effective in real-life situations, one could wear nothing but a sheet of aluminum foil, seal it tightly to minimize convective heat loss from cold air and feel warmer than if they were nude. However, this setup would actually make the person feel colder due to another scientific principle: the aluminum has a much higher thermal conductivity than air, meaning it would transfer the cold energy more quickly. **Fig.10** illustrates the thermal conductivity map of different materials.

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Fig.10 Thermal Conductivity Chart



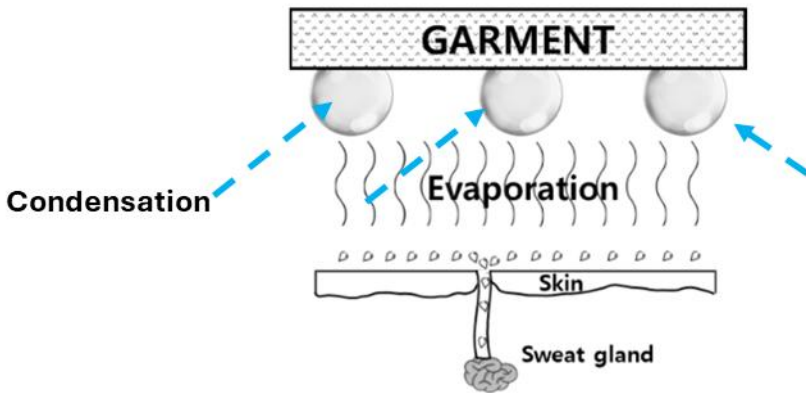
Dry air is among the least conductive, while metals are among the most conductive.

Basics of Thermal Science: In the thermal science relating to textiles, a critical area of focus is the microclimate, the region between the body and the garment system, as shown in **Fig.11**. This microclimate traps warmer air generated by the body heat. Furthermore, the degree of dryness (or humidity) within this localized microclimate creates a large impact due to heightened thermal conductivity of moisture and is determined by a complex interplay of factors, including the wearer's level of physical activity, health condition, and the thermal insulating properties of the textiles being worn. Thus, the more humidity in the area, the lower thermal resistance. Consequently, one of the most critical characteristics of thermal functionality in textile products lies in their ability to retain sufficiently large volumes of **dry air** layers between the body and the surrounding garment. In other words, the effectiveness of

thermal insulation is largely governed by how much dry air can be maintained within this microclimatic layer.

However, attempts to reduce radiative heat loss by inserting an aluminum sheet (or other solid materials) into this microclimate layer are unrelated to securing dry air layers and may have their intended effects easily negated, or even result in greater heat loss, due to the rapid conductive heat transfer caused by the high thermal conductivity of metals.

Fig.11 Microclimate Structure



Test: A study was conducted to measure the effect of the heat reflective technology on the thermal efficiency of a textile structure typically used for winter garments. The study compared the CLO values of a conventional non-woven thermal insulation material with and without a lining fabric printed with the silver reflective dots, obtained from a specific brand's garments promoting this technology.

About CLO: The CLO is a metric used to measure the thermal efficiency of a material in the apparel industry, similar to the R-value used in the construction industry. A lower R-value indicates a less efficient thermal protection, leading to higher energy costs for heating and cooling of a

house or building. Likewise, a higher CLO value in a textile material indicates a better thermal efficiency in conserving the body heat.

Test Environment: CLO tests are typically conducted in a controlled laboratory environment with a testing device, which includes a heat plate covered by the tested material, typically in a sheet form factor. The plate is heated to a certain temperature, for example, the body temperature of 36.5°C (97.7°F), and once the temperatures of the plate and the air above the material stabilize, the sensors located in the air above the material measure the heat energy trapped in the material by subtracting the temperature difference of the plate and the air, which is then expressed as a CLO value.

Results: The difference in the CLO values between the samples with and without the heat reflective liner fabric was less than 0.2 CLO, a range too small to be considered detectable by the general population. Given that the liner fabric hosting the reflective dots already has its own intrinsic CLO value, the impact of the reflective dots was negligible.

Real-World Scenarios: Under the premise of the heat-reflective technology, it might be more effective in cold weather when a person is engaged in high-intensity activities, causing the body to emit more radiative heat. However, in such scenarios, the body is already warm and doesn't need additional heat from the reflected energy. In fact, reflecting the heat back in these situations could do more harm than good by further raising the body temperature, increasing sweat levels in the microclimate,

and potentially leading to hyperthermia, as heat exchanges occur more rapidly in moist air.

However, consumers have no reason to be concerned about these issues. In general, when people consider typical winter outdoor activities and clothing choices, the mindset does not invoke extremely intense physical exertion or heavy perspiration without a plan for escaping from extreme weather conditions after the activity cases. In reality, after a vigorous physical activity, it is essential to move into a warm indoor environment as quickly as possible regardless of the clothing technologies or warmth level it provides. Failure to do so may result in a rapid drop in body temperature and lead to even a life-threatening risk. In other words, what consumers should be concerned about in such situations is moving to a space where the body can be warmed in order to eliminate this serious health hazard, rather than relying on thermal reflection effects that may prove to be entirely ineffective, which should be strictly avoided.

Based on the various scientific perspectives raised above, it is difficult not to harbor significant doubts regarding the reliability and real-world applicability of this thermal reflection technology, and I have not encountered any scientific theory or information to date that would adequately resolve these doubts.

Importance of Thermal Technologies: Unlike highly specialized garments such as flame-resistant suits for firefighters or bulletproof vests for police officers, most everyday clothing is chosen for more mundane factors like appearance, materials, comfort, and price, rather than for its protective features. However, winter clothing is an

exception, as it is essential for the health and safety of consumers in cold climates. While some may argue that a placebo effect in winter apparel is harmless if it provides a sense of warmth, in reality, relying on inadequate protection in cold weather can be dangerous and the potential harm should not be taken lightly.

Environmental Impacts: From an environmental and sustainability perspective, the printed patterns with different materials on the fabric surfaces introduce unnecessary impurities. While more details will be provided in **Chapter V: Solutions**, specifically in **Clean Recycling Initiative™** section, it is easy to recognize that if synthetic fibers account for three-quarters of global textile consumption, the success or failure of recycling within this material group has the potential to become one of the most effective and impactful environmental conservation efforts. Furthermore, the success of recycling efforts, particularly with synthetics, relies heavily on the material purity and the foreign materials used in the heat reflective technology can hinder the efforts.

1.4.2 Phase-Change-Material (PCM) technology

Another performance feature whose value I find difficult to comprehend is the Phase Change Material (PCM) technology, commonly applied also in winter textiles. The PCM technology involves certain materials that change their phases from, for example, liquid to solid, and maintain a constant temperature during the phase change. For instance, when ice melts, it transitions from solid to liquid state, maintaining the constant temperature of 0°C (32°F) during the process. Similarly, water maintains the

temperature of 100°C (212°F) during its phase change from liquid to vapor when boiling. A cooling gel used for stressed muscles maintains a consistent temperature while transitioning from semi-solid to liquid state on the skin.

Important Factors: The effectiveness of this PCM principle depends on several factors: the activation temperature, and the duration for which the material maintains a certain temperature. For example, to provide adequate protection in cold climates, the material must activate at a desired temperature and sustain it for a reasonable period. Therefore, evaluating the effectiveness of PCM in textiles requires an understanding of these factors. Unfortunately, many companies promoting PCM technologies often focus on marketing slogans like "*NASA used it for astronauts' suits on the moon*" rather than providing this crucial information.

Scientific Analysis: This misleads many people, who assume that, because the moon is extremely cold, this must be an effective technology. This is a typical example of how Artificially Conceptual Values (ACVs) can create misguided assumptions. However, for those with an analytical mindset, the issue is clear: the environmental conditions on the moon are vastly different from those on Earth, and the amount of PCM used in astronauts' suits is unknown. Our experience with ice gel packs for cooling sprained muscles provides insight into the importance of quantity. A gel pack of sufficient size and weight offers lasting relief, a concept which an astronaut's suit can easily accommodate in the effective use of PCM. However, if only a small amount were applied to a winter jacket as we cannot wear an attire as bulky and heavy as astronauts' suits, it would change its phase so quickly that the chemical

application would offer virtually no benefit. Therefore, just because a technology works on the moon does not guarantee its effectiveness on Earth in real-life.

Environmental Impacts: Typically, a PCM applied on textiles is a chemical material derived from petroleum sources, with the paraffin wax being the most common type. The material consists of long-chain hydrocarbons with 20 to 40 carbon atoms. As discussed earlier, long-chain hydrocarbons are used in water repellent treatments for garments. Certain water repellent chemicals such as C8 are banned in many countries due to their carcinogenic properties and long persistence in nature. Although these PCMs are designed to adhere to textile materials for extended periods, they gradually wash off during laundering and regular wear, contaminating the environment.

When purchasing textile goods containing PCMs, it is essential to understand the specific technical details mentioned above. From a performance perspective, a purchase should only be justified if the performance metrics align with shopper's intended needs. In practical textile applications, I have yet to observe any benefit from PCM materials in real-world situations. In most cases, PCMs are combined with conventional thermal insulation materials because they are either insufficient on their own or not effective at all, which poses a question for its usefulness despite of the associated environmental harm.

1.4.3 Heat-Generating Technology

One prominent example of ACVs involves a heat generating technology, often marketed with various other performance benefits, including several health benefits such as blood

circulation enhancement, negative ion emission for improved mood, etc.

Nature of ACVs and Analytical Skills of Consumers: After considering the earlier examples, readers might wonder why these apparent ACVs exist. In fact, there are many others that if I were to list and explain them all, this book would extend to additional hundreds of pages. Moreover, new ACVs will inevitably emerge as various organizations continue to find easy ways to appeal to consumers without making necessary efforts in truly effective technological advancements. I believe that the individuals and organizations making such claims often lack the depth of knowledge needed to fully understand the implications of their assertions. Unfortunately, this is a significant part of the dynamics of the consumer market today, once again highlighting the “unfortunate” nature of being a consumer.

Since it is impractical to cover all existing and potential future ACVs, the primary goal of this book is to help readers develop their analytical skills to better identify and recognize these misleading claims, an extendable skill for evaluating various environmental claims too.

Magical Technology?: Returning to the subject of the heat-generating technology, it can seem incredibly appealing, even almost magical, to believe that a piece of garment with the technology can offer such significant benefits, not only providing warmth but also improving blood circulation, mood enhancement, etc. Let us delve into the foundations of this technology.

Manufacturing Principles: Extremely small particles of various materials including ceramic, clay, carbon nanotube,

etc., are added to and evenly distributed in a polymer tank where the polymer is in a molten state. Fibers are then extruded with these particles embedded in them. According to the companies which promote this technology, these particles alter the properties of the fibers with the proposed characteristics, creating the purported benefits. To substantiate these claims, they frequently use seemingly scientific data obtained from specifically designed tests such as the one explained below.

Test Methods and Results: A common testing method involves comparing two fabric samples: one containing the purportedly beneficial particles and another without. These fabrics are placed at the bottom of a test chamber with a high-wattage light source positioned directly above them. The specific test I observed utilized a 500-watt Kawasaki lamp placed approximately 20 ~ 30 centimeters above the fabric samples. Temperature probes are inserted beneath the tested fabrics to monitor the temperature changes.

With the test setup complete, the light source is activated, and the fabrics absorb the heat energy from the bulb. The fabric containing the particles is claimed to exhibit a faster rate of temperature increase compared to the control fabric without the particles. In the observed test, a temperature difference of 10°C (18°F) was achieved within 30 ~ 40 minutes after the initiation.

Bases of Claims: The observed temperature rise in the fabric containing the particles is interpreted as evidence of the increased energy absorption from the light source. The more energy the material absorbs, the more it emits in the desired forms of energy such as infrared-, negative ion, etc.

This premise suggests that the garments incorporating this technology would function similarly during real-life use, absorbing energy from sunlight or body heat and subsequently releasing it to generate the advertised benefits. Companies making such claims may present additional supporting information, drawing analogies to related natural phenomena that supposedly produce similar effects. Examples might include experiencing health benefits while sunbathing or walking in a forest, etc. However, the validity of these analogies and the scientific basis for the claimed functionalities require critical evaluation.

Interpretations and Truth

Several key issues undermine the credibility of such claims regarding the enhanced heat generation and other associated functionalities.

Energy Level: A high-wattage light bulb generates an extremely high surface temperature and emits a level of energy far exceeding what humans can tolerate. For example, a 500-watt Kawasaki lamp becomes extremely hot, with a surface temperature easily reaching 400 ~ 500°C (752 ~ 932°F). At this temperature, it is impossible to put skin within 20 ~ 30 cm (8 ~ 12 in) distance of the bulb without immediate burns. This illustrates that the test conditions used to support these claims are not realistic. Companies may argue that such tests are stress tests, intended to demonstrate the effect under extreme conditions and that the results are proportionally applicable to lower temperatures, such as those from the sun or human body heat. To test this hypothesis, I conducted an experiment.

Experiment: I took two pieces of fabric samples, one containing the particles and the other without, to the rooftop of a 10-story building on a hot sunny day, setting them up on the floor. The only principal difference from the laboratory test method was that the heat source was the sun instead of the 500-watt Kawasaki bulb. Although the conditions were not exactly same, my test was much closer to real-life scenarios. The rooftop was extremely hot due to the scorching sun, with an ambient temperature of 30°C (86°F) with no wind. After the setup, I monitored the temperature change on the two fabrics every 10 minutes. No difference was found between them for the duration of two hours, then I decided to stop.

Related Science: It is important to recognize that stress tests are typically performed to study a material's behavior under accelerated and extreme conditions. These tests are inherently limited when it comes to directly applying the results to real-world scenarios under normal conditions because many materials around us exhibit non-linear behaviors. Assuming that the outcomes observed under a 500-watt Kawasaki lamp will occur in real-life situations is overly simplistic and scientifically inaccurate. It is clear that even the test conditions I used are extreme for most people, such as being exposed for two hours under the scorching sun at an ambient temperature of 30°C (86°F). Moreover, the energy level from the sun in my test is unlikely to be encountered during winter, when this technology is more relevant, while the energy level our body generates is nowhere near that of the sunlight, let alone a 500-watt Kawasaki lamp.

Health Benefits?: Another perspective on this claim involves its purported health benefits. In some forms of oriental medicine, infrared lamps are used alongside acupuncture to enhance the blood circulation of a patient and promote healing; An infrared emitting lamp with a typical temperature range between 200 ~ 300°C (392 ~ 572°F) is directed at the treated area with a certain distance and limited duration of exposure, where needles are typically inserted into the skin, supposedly amplifying the therapeutic effects. However, even when using infrared lamps in this manner, it is necessary to maintain sufficient distance to prevent skin burns, as the very assumption that health benefits can be achieved by exposing the human body to energy levels beyond what it can physiologically tolerate does not hold up scientifically.

Mechanics and Improbability: Examining the mechanics of this technology reveals further issues. The quantity of the particles used to create these fibers is limited because the molten polymer must be extruded through extremely small holes – more information on this extrusion process presented in **Chapter II: Synthesizing - Polymerization of Synthetic Fibers** section.

Adding the particles into a polymer melting tank is tricky because the small holes can be easily clogged up. Furthermore, the particles can settle to the bottom of the tank, which can create significant issues for continuous operations, such as fiber breakage and process failures. Typically, the particles do not exceed 10 ~ 15% of the weight of the molten polymer due to the concerns and process challenges.

Having had plenty of experiences in polymerization and extrusion processes in my career as textile engineer, I can attest that 10 ~ 15% is an extremely small quantity to create such magical benefits. Moreover, a large portion of the particles are encased within the fibers, surrounded by numerous polymer chains. For this technology to work, the sun's rays or the body's heat energy must penetrate the polymer structure, reach the particles, reflect enough energy, and then travel back through the structure to reach the skin. This scenario is, in my view, implausible even in science fiction.

If this technology truly had such remarkable benefits, one could simply bring a piece of rock into their living room and sit on it to gain the same effects, as the rocks naturally contain a much higher concentration of the substances claimed to provide these benefits. If it were that simple, this practice would already be widespread across the world.

Cautions and Environmental Impacts: Not to be persuaded by these types of ACVs, cautionary approaches are necessary. If a company presents test reports indicating medical effects, such as negative ion emission or far-infrared emission similar to the heat generation test results reviewed earlier, it is important to verify whether these tests were conducted under real-life conditions and whether the effects are proven to be truly beneficial for human health.

From an environmental perspective, the particles compromise the purity of the fibers and impede effective recycling, while the particles shed and contaminate,

thereby negatively impacting the environment during manufacturing and consumption.

I've highlighted several examples of ACVs and performance features that, when fully considered, offer little or no practical benefit while potentially harming the environment. In fact, I regard some of these technologies and practices widely practiced in the industry as truly absurd.

Historically, profit-driven organizations have often prioritized profits over social responsibility and due diligence. Even in my own professional experiences, I have witnessed many instances of similarly natured corporate misconducts.

Consumers often place too much trust in organizations around us, assuming they have thoroughly done their homework and offer products grounded with valid sciences. Unfortunately, time and time again, it has been proven that this trust may be misplaced and exploited. This underscores the vital importance of consumer education, a topic I will explore further in **Chapter V: Solutions**, in the **Education, Education, and Education** section.

1.5 Other Important Textiles and Environmental Impacts

In this subsection, I will explore a couple of textile technologies that can either greatly benefit or harm consumers while also having significant environmental impacts.

1.5.1 Home Textile & Energy Savings

The modern world demands unprecedented amounts of energy, with technology advances such as artificial intelligence and electric vehicles further driving up the energy needs. Consequently, the importance of conserving energy has never been greater. With this pressing issue in mind, I introduce a simple, low-cost solution involving a textile application that can significantly reduce the worldwide energy consumption.

Construction and Energy Efficiency: Considerable efforts have been made to enhance the energy efficiency of our living environments, including homes, appliances, commercial buildings and maintenance practices, etc. Many governments offer incentive programs to encourage their citizens to adopt the materials and technologies that improve the energy efficiency of their living.

One such example is the effort to reduce the rate of heat exchange between the interior and exterior of establishments. As briefly mentioned along with the CLO, the thermal efficiency, expressed as R-value, is measured and optimized on construction materials. Furthermore, thermal imaging techniques are used to identify the areas of high heat exchange for necessary improvements.

Windows and Energy Loss: A defining trend in modern homes and buildings is the increased use of large windows for an aesthetic appeal and better views. Unfortunately, windows are inherently less efficient at preserving energy compared to walls, which are typically insulated with specialized materials. **Fig.12** shows an example of excessive heat loss through window areas.

Fig.12 Heat Loss through Windows



<https://loe.org/shows/segmentprint.html?programID=12-P13-00046&segmentID=5>



In the image, the areas shown in orange indicate regions where insulation is insufficient and significant heat loss occurs, primarily corresponding to the window sections. In contrast, the areas shown in purple represent wall surfaces with more effective insulation. Even within these wall areas, the differences between insulated and non-insulated sections are clearly visible through variations in color intensity.

While many factors influence a building's energy efficiency, such as total window areas, efficiencies of air conditioning or heating systems, orientation of building and windows relative to the sun, integrity of window frames and seals, etc., some studies have shown that a significant portion of

energy, up to 70 ~ 90%, can be lost through the windows in thermally inefficient buildings.

Window Tint: Tinted glass windows can reduce heat gain during summer by blocking parts of sunlight, leading to lower cooling costs. However, this approach has its limitations. Many people value window transparency and are hesitant to darken their windows too much, which reduces potential energy savings. Additionally, tinted windows can block beneficial heat gain in winter, increasing heating demand.

Thermal Break: Some newer window frames are designed with two or more layers of glass, separated by gaps that act as a thermal break, helping reduce energy loss in winter and heat gain in summer. While this solution can improve energy efficiency, it is often costly, potentially requiring thousands of dollars to replace existing window frames, even for small homes. For large commercial buildings with older window frames, retrofitting can be nearly impossible. Although this technology can reduce energy bills in both summer and winter, its effectiveness is somewhat limited by the laws of thermodynamics: larger air pockets between the glass panes transfer energy more efficiently than smaller, compartmentalized pockets, making it less thermally efficient overall despite the high costs.

Other Means: Some people turn to solutions like window films or bubble wrap to save energy, particularly during winter in colder climates. However, these options require considerable effort for seasonal installation and removal with annually recurring costs while incurring environmental burden from the used and disposed materials.

Although the measures above can help conserve energy, combining them with, or even relying solely on, “thermally insulated curtains” can greatly enhance the thermal efficiency of homes and commercial buildings.

Thermally Insulated Curtains

The thermally insulated curtains are a cost-effective and simple solution suitable for any homes or commercial buildings, regardless of the age of construction or window size and location. These curtains are made with specifically designed synthetic non-woven fabrics, which are engineered to provide optimal thermal efficiencies while using a minimal quantity of textile materials.

During the manufacturing process of such non-woven fabrics, numerous tiny air pockets are created, forming an effective thermal barrier year-round. In summer, these curtains minimize heat gain from sunlight, creating a cooling effect, while in winter, they minimize heat loss from heating systems. Although heavy blackout curtains can provide similar benefits, optimally engineered thermally insulated curtains offer superior thermal efficiency, leading to a better energy conservation while using less textile materials.

These thermally efficient curtains can effectively address all forms of the heat exchanges in a building, namely conduction, convection and radiation. In general, the effects they create are far superior to any other methods mentioned earlier.

Commercial Availability: Despite their significant benefits, affordability, and flexibility, these thermally insulated curtains are difficult to find in the market. A USA retailer briefly offered them many years ago, but the product line was discontinued due to a low popularity and several logistical challenges associated with the product for the retailer. For example, these curtains took up considerably more spaces in their retail stores and warehouses compared to their non-insulated counterparts. Although this difference is negligible once installed in homes and buildings, the increased retail space requirement and associated challenges combined with a low popularity among consumers likely influenced the retailer's decision to discontinue the product.

Consumer Awareness: In my view, three main issues contributed to the unsuccessful market deployment of these curtains at the time.

First, had the consumers been made aware of the potential savings with hundreds or even thousands of dollars in electricity bills and significant environmental benefits, many more would have purchased the product. A comparable example is the widespread adoption of solar panels on homes and buildings, where homeowners are willing to invest thousands or tens of thousands of dollars while enduring significant renovation work during the construction of their living quarters.

Second, the increased storage spaces and handling costs for the retailer could have been easily offset by higher price tags. If consumers had understood the clear benefits, many would have been willing to pay a premium; Based on my industry experience and knowledge of textile goods' cost

structures, the additional cost to consumers for these products would likely be only a few dollars. Furthermore, the logistical challenges could have been addressed with a vacuum packaging, reducing the product's volume and making it comparable to, or even smaller than, other heavy blackout curtains. While vacuum packaging might slightly reduce the thermal efficiency of the product, a careful engineering can minimize this impact.

Third, at the time, already more than a decade ago, the public awareness and concern regarding environmental issues were significantly lower than they are today, and as a result, environmentally friendly slogans did not carry the same level of influence or effectiveness.

If any retailers, curtain brands, or related businesses are reading this and interested in exploring this concept, I would be more than happy to assist with product engineering, logistics, and even some marketing ideas. Based on this analysis, I believe the failure of the retailer's can be easily avoided in today's market.

1.5.2 Harmful Face Masks for Health and Environment

This topic concerns a particular type of mask product that is intended to benefit human health but, paradoxically, has the potential to pose a serious threat.

Even well before the COVID-19 pandemic, many people in Asia had recognized the importance of wearing face masks when facing respiratory viral infections due to several previous health crises like the avian flu, the H1N1 influenza, the MERS, etc. Additionally, masks were commonly worn as a protection against severe air pollutions in heavily industrialized regions. So, when the COVID-19 emerged,

people instinctively reached for masks in their closets or from local stores.

For many, it was surprising when Dr. Anthony Fauci, the head of the National Institute of Allergy and Infectious Diseases (NIAID) in US, initially stated that masks were unnecessary for the general public. He emphasized the importance of reserving masks for healthcare workers and suggested that while masks might offer some protection, they were not essential for those who were not infected or caring for the infected.

Then, several weeks later, the guidance was reversed, and the public was urged to wear masks during all social interactions. While there is an ongoing debate, albeit by some minority opinions, about the effectiveness of face masks in preventing the spread of respiratory viral infections and, also acknowledging my lack of expertise in the medical science, I refrain from drawing any conclusions about whether wearing face masks is right or wrong in this discussion. However, based on the conventional understanding promoted by the authorities around the world after a short period following the breakout and other similar viral outbreaks of the past, the initial response from Dr. Fauci appears to have been mistaken.

Importance of Developing Analytical Skills: In my view, prioritizing healthcare workers' protection was something most of our society would have readily supported. However, providing accurate information to the public is equally critical in the efforts to mitigate the crises of this nature. The initial lack of understanding in some countries about the asymptomatic transmission of the COVID-19 stood in a

stark contrast to the immediate mask-wearing response in many Asian countries.

This incident highlights that even top authorities can make mistakes. In fact, there have been many other examples of such occurrences, some of which will be discussed later in this chapter and in **Chapter V: Solutions**, specifically in the **Section 20. United Efforts**. These examples remind us that blindly following authorities may not always lead to the best outcomes, especially when it comes to matters as crucial as our health and safety.

While I do not aim to promote conspiracy theories or encourage those who spread misinformation and provoke unwarranted fear, we must develop analytical skills needed to recognize flawed guidance. In such situations, it is crucial to approach the issues with a scientific and analytical mindset and, at times, rely on common sense. While expert advice is invaluable, applying critical thinking and questioning authority on questionable guidance is key to managing our lives more effectively.

At this point, I would like to make it clear that I have absolutely no intention of defending those who promote conspiracy theories or spread misinformation that unnecessarily incites fear. What I wish to emphasize is that cultivating analytical and critical thinking skills that enable individuals to discern inaccurate information is an essential capability that each of us must possess in order to navigate our lives responsibly. Disregarding expert advice is undoubtedly a dangerous way to live; however, as illustrated by the mask-related controversies in the United States discussed above, there are areas in which analytical and critical thinking are indispensable, not only for

protecting our own health, but also for advancing environmental protection.

Copper Particle Masks

The lengthy prelude to this subsection serves to alert people about a health product that poses significant risks.

Introduction of Dangerous Mask: Recently, I was in an online meeting with an Asian manufacturer regarding a specific technology I was seeking. During the call, I learned that the company was involved in various other businesses, including face masks. After completing the discussions on the planned subject, the manufacturer began promoting a specific mask product, which featured copper nanoparticles. As they showcased a sample on the camera, I noticed a diluted yellowish tint evenly distributed across the mask, indicating the presence of copper in small particle forms.

From a particular research project during my tenure as research and development engineer in South Korea, I gained considerable insights about the concept behind this product: the copper particles are intended to kill microorganisms by releasing electrically charged ions that destroy viruses and bacterial cells in contact. Silver has similar properties, but it is much more expensive. The manufacturer claimed that these masks were quite popular with customers in many countries including USA, Japan and many countries in Europe.

Significant Concerns and Flag: However, my experience in the project in South Korea immediately raised serious concerns on this mask product. Metal particles, such as

copper and silver, cannot be permanently fixed to fibers and fabric structures. In fact, a significant quantity of these particles is released into the air during use, and the release quantity depends on how they were impregnated into the material structures during manufacturing.

Research Project: In the research project, I was tasked with developing acrylic fibers impregnated with copper particles to reduce static electricity that people experience when putting on or removing acrylic sweaters. The idea was that the metallic ions in copper would help dissipate the stagnant electricity. From that experience, I learned that, despite my best efforts to keep the process clean, copper particles contaminated the entire laboratory surfaces. When I swiped my fingers across any surface, I observed a greenish-yellowish tint, typical of the copper particles in high concentration, on my skin.

Danger and Potential Health Impacts: Placing such particles in the path of our breath is dangerous. It is not difficult to imagine that these particles can be easily inhaled through the mouth and nose while breathing. Once inhaled, they can reach organs and bloodstreams in the body. For instance, asbestos, now banned in many countries after extensive research, causes lung cancer because its fibrous particles scratch and damage lung tissues. Metals like copper, being potentially harder than the asbestos, could inflict even more damage inside the body. Additionally, metals can trigger other unknown reactions due to their heightened chemical reactivities.

After my meeting with the manufacturer, I searched the ChatGPT, which resulted in the following description:
“During the COVID-19 pandemic, the FDA issued

Emergency Use Authorizations (EUAs) to facilitate the availability of various types of face masks, including those with antimicrobial materials like copper and silver, to help prevent the spread of the virus.”

Although I immediately advised the manufacturer against selling the mask product during the meeting, they likely believed that the FDA’s approval carried higher authority and validity than my anecdote and assessment for the health risks.

Variables and Environmental Impacts: If a mask is made with copper or silver in a strong filament yarn form factor, it could serve its intended purpose as antimicrobial with a lower health risk, as the continuous filaments are much less likely to shed particles during use. However, the masks made with copper or silver in such form factors as particles or fibers significantly increase the risk of the particle inhalation. In my research project, I embedded fine copper particles into molten acrylic polymer in polymerization process and then extruded it into fibers. This process is one of the most secure methods of embedding fine particles into various forms of polymer structures. However, even with this secure method, the resulting fibers still contained considerable quantities of copper particles on their surfaces, which easily transferred onto my skin when touched.

Readers, please do not wear the face masks made with any particles regardless of its kind. Inhaling the particles can pose more harm than any potential benefit.

Environmentally, small particles of metals with their inherently heightened chemical reactivity will persist in the

environment and create significantly harmful effects on all living organisms.

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Section 2 “Fast Fashion”

Although this topic aligns with the theme of the previous section, **Unfortunate Consumers**, I felt it necessary to create a dedicated section due to the widespread misconceptions surrounding it and its importance in key environmental initiatives.

“The term, “fast fashion” was first popularized in a New York Times article in 1989 to describe retail store Zara’s first opening in the United States” – CNN

<https://www.cnn.com/style/what-is-fast-fashion-sustainable-fashion/index.html>



Since then, the concept has rapidly gained global attention and is now frequently used as a symbol of various environmental issues associated with textile goods and organizations, often based on unscientific and vague notions.

Disagreement: The term "*fast fashion*" is encapsulated in the quotation marks throughout this book, reflecting my disagreement regarding its conventional usage and definition. This reservation is rooted in my background in textile engineering and decades of industry experience, which provide me with a deeper understanding of the complexities of textile production, quality, and market dynamics than most in the public, media, environmental advocacy groups and activists, including the New York Times reporter who created it.

Varying Perceptions: The ambiguity surrounding the concept of "*fast fashion*" is evident in the varied public

perceptions of different apparel brands in the market. Some consumers may categorize a particular brand as "*fast fashion*," while others may not, highlighting the subjective nature of this classification. These divergent opinions often arise from the differing interpretations of various aspects of business operations, which may be motivated by the prices of the products in stores, the size of them and how flash they look. However, all these factors may not necessarily relate to the quality or durability of the goods categorized as "*fast fashion*".

As a habit of a person, deeply involved in the textile industry, I often conduct informal surveys, asking people around me on various subjects related to textiles. One such example is to ask them to classify well-known brands such as Lululemon, Adidas, Nike, etc., as either "*fast fashion*" or not. The responses I receive are notably diverse, with no clear consensus emerging for any particular brand. This variability in people's perception extends even to some brands, which are considered as "luxury" brands by a large portion of the public, primarily due to the high retail values of their merchandises, surprisingly classified by some as "*fast fashion*".

Uneducated Notions: Given the vagueness, the public might recognize that the original coining of the term "*fast fashion*" in the 1989 New York Times article may have been based on uninformed notions by the author, rather than a clear and scientific understanding of the complexities of textile quality.

In light of this, I present several points for consideration concerning this situation in the following discussions. These deeper examinations will reflect my view of the term

as more of a social phenomenon than the accurate measure of textile quality and durability, underscoring the importance of a more nuanced understanding of textile materials and their environmental impacts.

2.1 Ambiguity in Definition

From the perspective of the New York Times reporter, all garments in Zara's stores would be categorized as "*fast fashion*," regardless of material choices, durability, or other factors related to the quality. This label comes with several associated perceptions and environmental concerns including:

Perceptions

- *"Fast fashion" goods are produced quickly and have shorter lifecycles due to poor quality.*
- *"Fast fashion" goods are made with cheap materials - a notion frequently associated with synthetic fibers*
- *"Fast fashion" goods persist in nature for too long - a notion once again associated with synthetic fibers*
- *"Fast fashion" goods do not last long in consumption*
- *"Fast fashion" goods create microplastic pollution from synthetic fibres*
- *"Fast fashion" goods lead to overconsumption*
- *"Fast fashion" increases carbon footprint due to reliance on air shipments rather than ocean transport*

Furthermore, the opposite concept is often described as “*slow fashion*” and represent more favourable images consumers form towards certain materials, merchandises, brands, forms of operations, etc.

Expanded Interpretations: In addition, the term “*fast fashion*” has acquired connotations beyond its original meaning, often encompassing broader social aspects of a company's operations. For instance, some individuals automatically associate overseas production utilizing lower labor costs with “*fast fashion*”, regardless of the actual quality of the goods produced.

Conversely, the varying interpretation of “*fast fashion*” appears to blur the boundaries between the product-specific characteristics in quality measurements and the broader concept of ESG (Environmental, Social, and Governance) practices within a company's overall operations. While some may argue that this broader definition provides a more comprehensive representation of a company or brand, there are drawbacks to such an expansive interpretation, particularly, in the perspectives of making more effective environmental efforts.

Mixed Values and Discrepancies: For instance, the quality, durability, and speed of factory-to-market for a particular product can be managed through specific operational strategies, which may be distinct from an organization's commitment to social and environmental responsibilities. If a company aims to align with consumers’ notion that synthetic fiber products are inferior in quality and longevity, they might simply substitute natural fibers for synthetic ones. The perceived improvement in the longevity from the switch would lead equally to the perceived improvement in

sustainability and this action can be executed independently of the company's efforts to fulfill its other responsibilities.

Combining these diverse concepts under a single term, therefore, may impede the public's ability to accurately evaluate matters related to the "*fast fashion*" and its overall environmental impacts. The following subsections will explore how ambiguous notions of the "*fast fashion*" can potentially lead to unintended consequences for environmental and sustainability efforts.

2.2 Manufacturing Factors – Material, Quality and Cost

This subsection begins by exploring the concept of "Quality", a term often misunderstood in the context of "*fast fashion*."

Definition of Quality – Oxford Languages

- *the standard of something as measured against other things of a similar kind; the degree of excellence of something.*
- *a distinctive attribute or characteristic possessed by someone or something.*

The definition emphasizes the "measurable" attributes or characteristics, which contrasts with ambiguity often associated with the products or brands labeled as "*fast fashion*". Therefore, using specific metrics commonly applied in evaluating textile quality provides a more scientific basis for assessing the true quality of the materials and goods associated with "*fast fashion*" and "*slow fashion*".

While "quality" can refer to many different factors, the following analyses focus on the material aspects of natural and synthetic fibers, as well as their derivatives like yarns, fabrics, and finished products. For instance, if a seamline breaks, it is more likely due to poor workmanship or the weakness of the sewing thread, rather than the fabric's strength. Although a consumer may interpret this as a sign of "poor" quality, this discussion distinguishes the scientifically measurable durability of the fabric from other potential causes of failure. Here, I present these quality factors with a focus on the material considerations.

Strength: Synthetic fibers are generally five to ten times stronger than natural fibers - A detailed analysis of this topic will be presented in the section, **3.2.1 Strength Factor and Dust Quantity**, with supporting scientific data. This strength advantage means that products made from synthetic fibers are better able to withstand forces such as abrasion, tensile stress, tear strength, stretching, etc., key factors that contribute to the failure of textile goods. The higher strength of synthetic fibers directly enhances the durability of the resulting materials, including yarns, fabrics, and finished products, throughout their entire lifecycles, compared to those made from natural fibers.

Durability Factors: While this is a straightforward scientific fact, it can become more complex when different care methods are applied to different items, as previously reviewed in this chapter. In a brief recap, home laundry involves water usage and significant physical forces from the washing mechanisms, leading to accelerated material fatigue and degrade whereas drycleaning does not involve such forces. Additionally, interactions between the

chemicals contained in textiles, water and detergents can increase the risk of color fading. Although synthetic materials generally withstand these physical and chemical forces exceptionally well, comparing home-washed items with dry-cleaned ones is not an accurate measure of their overall quality and durability.

Contradictory Perceptions: Furthermore, contradictory perceptions exist regarding "*fast fashion*" goods: "*They do not last long in consumption, but last too long in nature.*"

Scientifically, if a material lasts long in nature, it typically reflects a quality that also offers longevity in consumption. Therefore, an analytical mind can draw a conclusion that the perception, contradictory to the underlying science, may be created by different metrics such as varying measuring sticks with subjective quality references. Conversely, there is a significant gap in the public perception of quality and designations of certain brands as "*fast fashion*" or "*slow fashion*."

I recognize that many readers may not yet grasp the full extent of the scope presented so far in this discussion. It is natural as textile quality and manufacturing are more complex than most people think. With the tangible examples in the following discussions, the scope will become clearer.

Complex Manufacturing and Market Structures

Throughout my career as textile engineer, I have either worked in or visited countless factories involved in every stage of textile manufacturing, from raw materials to

finished goods. These experiences have shown me that it is not uncommon to find a single factory producing goods for both "*slow fashion*" and "*fast fashion*" brands simultaneously, using identical raw materials, craftsmanship, and processes.

Similar Qualities with Different Prices: These goods are essentially of similar quality, if not identical, but their price tags can differ considerably based on the brands in question and their retail strategies. For instance, one brand may sell its products in an upscale storefront, while another sells online with large differences in the prices between the two. Without knowing the origins of these goods, many consumers may distinguish them as either "*fast fashion*" or "*slow fashion*", "good quality" or "poor quality". This example illustrates the complexities of manufacturing and market structures which most consumers do not fully understand, yet it can lead to improper judgements on the quality of goods, leading to actions which impact environmental sustainability.

Different Qualities with Different Prices: Different scenarios do exist, which correspond more to consumers' general expectations for a higher quality in relation with a higher price in retail. For instance, a brand implements rigorous quality control programs to ensure that their products respect safety and quality standards for its workers and consumers. This may include avoiding harmful chemicals, which can be otherwise used to enhance visual appeal or reduce production costs, among many other operational measures taken by brands and manufacturers. Such approach requires a system with necessary resources, such as experienced personnel and effective operational

structures. This responsible approach adds to the cost structure of the organization, which is naturally reflected in the pricing of their products.

However, it is essential to recognize that certain operational systems and cost structures, while critical, should not be confused with the material quality of the products themselves in the context of “*fast fashion*” vs. “*slow fashion*” and its associated misperceptions described earlier. This distinction leads to a broader discussion on the nature of material quality and cost as follow.

Materials Matter: To illustrate the perspective of material choices, associated with their impacts on quality and costs, I use a scenario involving a brand using silk material for producing a dress instead of polyester, leading to a significant cost increase. Many consumers would prefer an expensive silk dress if they could afford it, not considering it as a “*fast fashion*” item. However, in terms of the “measurable” strength and durability, an equivalent dress made with polyester offers much higher strength, thus would last longer under identical care conditions. The silk dress could be ruined in a single heavy home laundry cycle, whereas the polyester dress would endure repeated washes.

2.3 Retail Price Factors

Among consumers, there is a common belief expressed by phrases like “*you get what you pay for*”, suggesting that higher prices equate better quality. This subsection explores the potential disconnect between this perception and the actual impact of material costs on the retail prices of various textile goods.

Cost Bases: Revisiting the example of silk and polyester dresses, it is evident that a portion of the cost is tied to materials, such as considerably more expensive silk yarn for the silk dress compared with polyester yarn for the polyester counterpart. However, numerous other factors contribute also to the final retail price. In fact, several other key factors exert a stronger influence on the cost structures, which in turn affect retail prices. As mentioned earlier in the discussion on **Different Qualities with Different Prices**, some costs are related to quality assurance. Of all associated factors in manufacturing, however, material costs are invariably a relatively small portion of a product. For example, in some local markets in China or Vietnam, beautifully crafted scarves and dresses made of 100% silk are sold at prices that are almost unimaginable for western consumers. While this can, of course, be attributed in part to the geographical advantage of proximity to the production site, enabling direct sales with minimal distribution costs, such costs in fact account for only a very small portion of the overall cost structure of a luxury-brand silk product. In other words, material and distribution costs are not sufficient to explain extremely high price tags on silk goods offered by luxury brands.

Other significant factors are non-manufacturing related, such as the point of sale (POS) structures and administrative functions. It is widely understood within the industry that a substantial portion of the costs associated with retail products comes from administrative operations, often far exceeding manufacturing costs, including materials. These administrative costs include employee and executive salaries, benefits, office space, storefronts,

inventory storage in major cities, and marketing expenses for advertising campaigns.

Meanwhile, the retail price of a given product is formed within an operational system so complex that it is difficult to fully understand without closely examining a company's internal cost structure; nevertheless, certain patterns are widely recognized within the industry. For example, even when discounts of 30 ~ 50% are offered during major sales events such as Boxing Day or Black Friday, sellers often still generate profits on those products. Furthermore, it is widely understood that by the time discounts of up to 70% are applied at the end of a season, manufacturing and administrative costs have already been fully recovered, and the product has either reached or come very close to its break-even point.

Although all of the information presented above is based on generally accepted industry discourse, it nevertheless allows us to infer the following general operational principle. Namely, the retail price assigned to a given textile product reflects a cost structure in which production labor costs exceed material costs, and administrative costs exceed production labor costs. When this framework is applied, it becomes clear that, for most textile products we purchase, material costs account for only a very small proportion of the final retail price.

Consequently, even when the raw material quality of silk scarves or dresses sold in local markets in China or Vietnam does not differ significantly from that used by ultra-luxury brands, an incomparable disparity emerges in their actual retail prices. In fact, because luxury brands also source their silk from these well-known regions that are globally recognized for high-quality silk production, it is entirely

possible that the material cost of a scarf purchased locally for five dollars and that of a one-thousand-dollar scarf beautifully displayed under the soft lighting of an elegant boutique are nearly identical.

Pricing Strategies: Additionally, pricing strategies vary across companies. Some brands may opt for lower sales volume with higher price points to maintain a desired brand value. Conversely, other companies may prioritize maximizing profits by selling larger quantities at lower margins. These strategies are often influenced by factors such as existing brand values, target markets, demographics, among many others.

Disparities – Quality and Price: Regardless of the rationale behind such price variations, a significant gap between raw material costs and retail prices often exists across both expensive and affordable textile goods. This disparity can result in consumers paying a premium for perceived brand value or administrative overhead rather than the inherent quality of the products.

Personally, I prioritize the value proposition of the goods I purchase rather than assuming a direct correlation between brand recognition and product quality. This approach involves a comprehensive evaluation that considers price, quality, durability, maintenance costs, and overall performance.

For instance, a suit I purchased from Zara, the brand initially labeled as "*fast fashion*" in the New York Times article, has remained in excellent condition for over 15 years. Contrary to the reporter's perception, I see no reason to associate it with the negative connotations of "*fast fashion*" in terms of

its quality. Despite avoiding highly recognized brands and expensive prices, I rarely fail in my purchasing decisions, even considering the environmental sustainability aspects of the items I buy.

At this point, there is one matter I would like to make particularly clear to readers: my arguments should not be interpreted as a defense of low-priced products. People have valid reasons for preferring products from specific brands, such as personal taste in materials, design, style, or brand image, and they are fully entitled to enjoy the personal sense of happiness that comes from pursuing such preferences. What I wish to emphasize is that these subjective considerations should not obscure or distort a scientific understanding of environmental impacts, nor should they lead it in a misguided direction. This perspective has accompanied the book from its very beginning, and I wish to reaffirm that it will remain central through to its conclusion.

2.4 Misperceptions and Overconsumptions

Apart from the quality and longevity perspectives presented earlier, labeling certain brands as "*fast fashion*" can inadvertently create misperceptions, leading to undesired environmental consequences by influencing consumers' overall consumption habits of textile goods.

Emotional Values: Most people, including myself, have certain belongings that they feel more attached to than others. They can be expensive items like houses and vehicles, or special gifts attached with precious memories.

Conversely, special attractions and perceptual values can certainly exist in textile goods. Personally, I value clothing that is comfortable to wear and easy to maintain, which stems from my background as textile engineer. This perspective may differ from that of the general public, who might not assess textile goods through the same level of technical lens as me. However, like others, I also attach emotional value to certain items, especially those associated with meaningful moments or gifted by special people. These items are often more cherished and cared for due to the emotional connections they hold.

Lower Values: On the other hand, some items may be regarded as having lower value by many. For example, an inexpensive synthetic shirt displayed in a store associated with "*fast fashion*" may lead a shopper to make an impulsive purchase. In this situation, the shopper may develop preconceived notions about the item's quality and durability, often aligning with the common perception that "*fast fashion* products have shorter lifespan".

Consumption Behaviors: Relating to emotional values and purchasing decisions, it is natural for most consumers to think more carefully when buying higher-priced items. The more expensive the product, the more planning and consideration go into the decision, which in turn creates an emotional value in itself. For this reason, people often vividly remember where and when they bought an expensive silk dress, while they may not have such clear memories of purchasing an inexpensive shirt on impulse.

Differences of Fates: Perceptions also influence how people treat items that show signs of wear. If someone

believes an item isn't meant to last, even a minor flaw, like a loose button or a broken seam, can prompt them to dispose of or donate the item rather than repair it. In times when materials were scarce, such issues would have been resolved with simple repairs, no question asked. Today's picture in wealthy countries, however, presents a stark contrast. Large quantities of textile items placed in community recycling bins or donated to charity groups often require only minor repairs. It is evident that the individuals who disposed of these items had little or no emotional attachment and valued them too little to retain them with small fixes.

In summary, labeling textiles as "*fast fashion*" can create a perception of lower value in consumers' minds, which may not accurately reflect the real quality or the longevity of the goods in question. This perception shapes the fate of these items, often leading to premature disposal in landfills or incinerators and subsequently driving the consumption of other textile goods as replacements.

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Section 3 Microplastic vs. Natural Fiber Dust

Dust, by definition, consists of small particles detached from their original body mass, either from natural or synthetic materials, due to various physical or chemical impacts, then spread primarily through air. Many individuals suffer from allergies caused by pollens during flowering seasons. Pollens are an example of dust originating from nature, illustrating that not all natural sources are harmless. The impacts of dust particles on health and environment can vary widely depending on two main factors: chemical compositions and quantity.

Since the invention of the plastics over a century ago, these materials have become deeply embedded in our daily lives. Due to their thin and long form factors, synthetic fibers have been a major source of microplastics for more than 100 years, shedding dust particles during manufacturing, normal wear and washing.

This contamination has led to growing concerns about microplastics in virtually all our food sources, and rightfully so. It is well known that microplastics are present in almost everything we eat and drink, even in unexpected places like inside meat, fish, and vegetables. These tiny particles are so small that they even float in the air we breathe.

In this discussion, I delve into the nature of the dust particles from textiles and their environmental and health impacts.

Natural Fibre Dust and...: We would be living in a rather strange, and almost magical, world if only synthetic fibers

shed dust (microplastics), especially considering that natural fibers are generally much weaker than synthetic ones - a point that will be explored in more depth in subsequent discussions later in this section. Yet, the dust from the natural fibers is largely overlooked. As a result, many readers may be unaware of its nature and impacts.

To address this topic on a scientifically fair basis, I will examine the sources and impacts of the dust from both synthetic and natural fibers, beginning with the scientific understanding on the microplastic.

3.1 Microplastics and Synthetic Fibers

Microplastics are generated from plastic sources through physical impacts during manufacturing and consumption. A common misconception is that microplastics mainly originate from the obvious plastic materials we encounter routinely in our daily lives like bottles and bags. While these sources do contribute, one of the most significant sources of the microplastics is the synthetic fibers in textile goods. This is due to their extreme length directionality and inherent weakness along the length side.

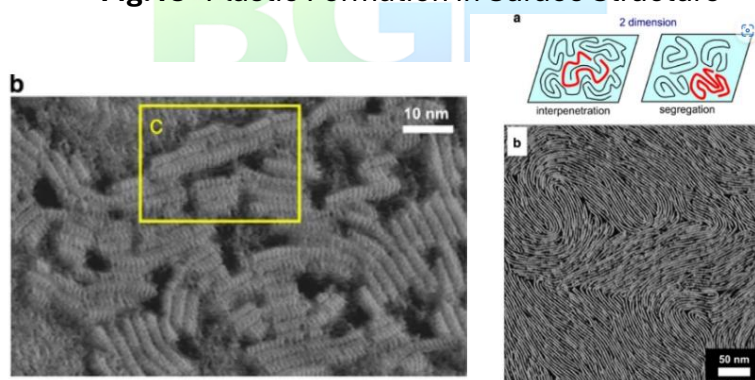
To achieve a more scientific understanding of this phenomenon, we will examine the structural characteristics associated with different forms of plastics and consider how these characteristics relate to the mechanisms underlying microplastic generation.

Forces for Microplastic Creations against Polymer Formation Structures: Scientifically, there are 13 different physical forces that can cause materials to break, and in the

case of plastics, this breakage often results in the creation of what is commonly referred to as "microplastic". These forces include tensile, compressive, shear, bending, abrasive, and frictional forces, among others.

When it comes to the structural integrity, which plays a crucial role in determining the amount of dust produced, plastic bottles and bags offer much greater resistance to physical impacts compared to synthetic fibers. This is due to their more favorable formation structures and form factors, which become apparent when examining their surface structures. Under extreme magnification, modern microscopes reveal that the polymer chains in plastic bottles or bags are tightly intertwined, providing strong structural integrity, as shown in **Fig.13**.

Fig.13 Plastic Formation in Surace Structure



<https://www.nature.com/articles/pj201567>



In this robust structure, the polymer chains are arranged in random directions, making the material inherently resistant from all directions of the physical forces mentioned earlier. Weaknesses arises, however, when plastics are formed into

thin sheets, like grocery bags, which are more susceptible to stretching, tearing and puncturing compared to thicker plastics, such as food containers.

Despite this, thin plastic bags still demonstrate an impressive strength-to-weight ratio, often capable of holding thousands of times their own weight (typically a 1:1,000 to 1:2,000 weight ratio). All readers of this book have likely experienced carrying groceries weighing several kilograms in a very lightweight plastic bag. This common experience makes it easy to appreciate the exceptional strength that plastic bags provide relative to their unit weight. Moreover, this robust structural strength plays a significant role in markedly reducing the likelihood of microplastic generation when compared to the synthetic fiber material groups that will be discussed next. This combination of strength and lightness has contributed to the widespread popularity of plastic bags and bottles, while significant environmental concerns surrounding them have grown.

Microplastics from Synthetic Fibers: In contrast, synthetic fibers are far more susceptible to breakage due to the extreme directionality created by the long and thin form factors. **Fig.14** below shows multiple strands of synthetic fibers observed under high magnification using a microscope. Each individual fiber strand has a thickness comparable to, or even finer than, that of a spider's web, and as a result exhibits pronounced mechanical vulnerability along a specific direction, namely, the longitudinal axis.

Fig.14 Fiber Alignment in Lenth Direction in a Yarn



The difference in strength becomes evident when comparing the two: while a thin plastic bag weighing 5 grams can hold 5 ~ 10 kg (11 ~ 22 lbs) of groceries, synthetic fibers can break under just a few grams of weight, which explains why a large number of them are bundled together and twisted into a yarn in part to enhance the strength – more on this in **Chapter II: Yarn-Making** discussion.

While the resulting yarn becomes considerably stronger, the comprising fibers remain weak. As a result, mundane everyday activities such as putting on or taking off garments, friction with other objects, or laundering can easily generate enough force to break them and produce microplastics, although such microscopic phenomenon cannot be seen or recognized with our senses.

A study by a team at a reputable university highlights the significant amount of microplastics released during laundering. They reported that an average washing load of 6 kg of laundry could release an estimated 137,951 fibers

from polyester-cotton blend fabric, 496,030 fibers from 100% polyester, and 728,789 fibers from 100% acrylic.

Cautionary Analysis: While the data clearly indicates the generation of a large quantity of microplastics from a single laundry load, I approach these findings with caution. Three key considerations warrant this careful analysis: First, the methods used to capture microplastics, as even the most advanced filtering systems cannot capture all microplastics due to their extremely small and lightweight nature. Second, it is extremely difficult, if not practically impossible, to collect all particulate matter generated from both natural and synthetic materials during laundering and then accurately separate and quantify the respective contributions of each material. This limitation is particularly evident in experiments involving polyester–cotton blends, where such methodological constraints may compromise the accuracy of the reported data. Third, the study fails to provide even the most basic information that should be considered essential. For instance, in the case of polyester–cotton blended fabrics, the blend ratios of each material are not disclosed, nor is there any information regarding the dust generated specifically from the cotton fibers. Likewise, for 100% polyester and 100% acrylic fabrics, the study does not specify the denier of the fibers used.

For these reasons, I approach such research materials selectively, distinguishing between aspects that can be meaningfully adopted and those that should be treated with caution or discarded. I do so because I am well aware that uncritically accepting such figures, simply because they are presented by reputable institutions or authoritative figures, does not contribute to the development of my own

understanding, nor does it meaningfully advance the cause of environmental protection.

3.2 Dust of Natural Fibers

Similar to synthetic fibers, natural fibers exhibit extreme directionality with the thin and long form factors. The same physical forces that generate microplastics from the synthetic fibers, such as tensile, shear, and frictional forces, also apply to the natural fibers.

The following discussion provides an opportunity to examine, in a comprehensive manner, the quantity of particulate matter (dust) generated from natural fibers, the chemical substances contained within them, and the resulting environmental impacts.

3.2.1 Strength Factor and Dust Quantity

To comprehend the quantity of dust generated from textile materials in both natural and synthetic compositions, it is essential first to grasp the concept of ‘intrinsic strength,’ which quantifies the level of resistance each fiber type exhibits against applied physical forces (loads). In this discussion, we employ the breaking energy analysis method, explained in the *****Industry Specifics***** section below.

As stated in the preface of this book, the content in this section is intended to offer more detailed technical information for professionals working in the relevant industries as well as for students studying textiles or related fields. General readers may choose to skip this section without difficulty in understanding the overall argument or

context of the book. That said, certain elements within this section may be important for all readers to grasp, particularly the information contained in figures. These will be sufficiently explained in the subsequent discussions, and the book is structured such that readers can follow along naturally, referring back to the relevant figures or information as needed without encountering difficulty.

*****Industry Specifics (1) – Start*****

Test Method: Each fiber strand is affixed to two gripping mechanisms, one at the top and one at the bottom, within the test equipment, illustrated in **Fig.15**. The top gripping head then pulls the fiber strand in the length direction until it breaks. During this process, the sensor attached to the moving head measures the breaking force and the distance the fiber travels until breakage. This method, commonly referred to as “tensile test”, is also applicable to various other intermediary textile materials such as yarns and fabrics - It should be noted that the sample shown in **Fig.15** consists of a yarn, not a fiber strand. Beyond the textile applications, it is often used in such materials as films, steel, etc., where tensile strength is a critical quality indicator.

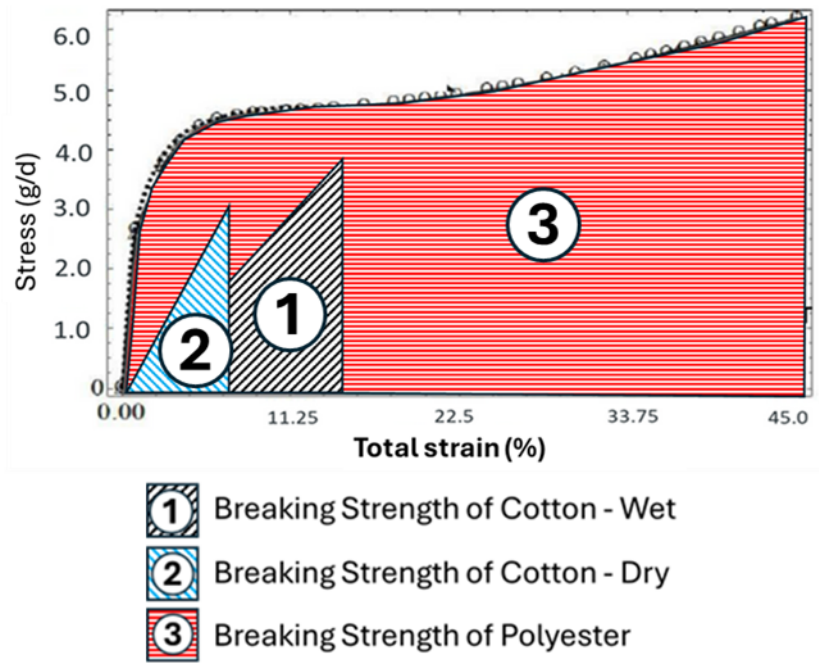
Fig.15 Tensile
Strength Tester



In standard laboratory practices, a selected number of fibers are tested based on a statistical method for determining the sampling size, and the average values of related properties are reported.

Stress-Strain Curve: **Fig.16** provides an example set of data obtained from a tensile test. The value on the Y-axis represents the stress, and the value on the X-axis represents the strain, hence the name "Stress-Strain (SS) curve."

Fig.16 Breaking Energy Measurements



Fiber	Tenacity (g/denier)		Elongation (%)	
	Dry	Wet	Dry	Wet
Cotton	2.4 ~ 2.9	3.1 ~ 3.6	7 ~ 9	12 ~ 14
Polyester	4.8 ~ 6.0	4.8 ~ 6.0	44 ~ 45	44 ~ 45

The SS curve is highly useful for a wide range of analyses on fiber characteristics as it offers comprehensive insights into the physical behaviors and internal structures of the tested materials. Given the scope of this book, I will not delve deeply into the technical analysis on this particular set of data. Instead, I will focus on analyzing the properties related to the dust creation between polyester and cotton.

Analysis: The SS curves for the cotton and the polyester reveal distinctly different patterns. The cotton curves typically form triangles, while the polyester exhibits curved shapes with multiple inclinations before breaking. The shape of the polyester on **Fig. 16** reflects its internal polymer structure characteristics, particularly its degree of crystallinity, a trait shared with other synthetic fibers: the initial steep slope of the polyester indicates the strength required until the tensile force reaches a yield point, the point at which the crystallinity no longer provides the dimensional strength to maintain the original fiber structure. Beyond this point, the curve's slope decreases, showing that the fiber stretches more easily with the applied unit force. The degree of the stretchiness is the function of the non-crystalline parts of the polymer structure. Together, these characteristics contribute to the total energy required to break the fiber.

In contrast, the cotton exhibits a straight line until it breaks, a typical behavior for most plant-based fibers. This can be largely attributed to the cellulosic structure of cotton, which has a more uniformly distributed structure throughout and lacks the combination of the distinct crystalline and non-crystalline regions found in synthetic fibers.

*****Industry Specifics (1) – End*****

The key point that readers should understand in this analysis is that the area under each graph labeled **(1)**, **(2)**, and **(3)** in **Fig. 16** represents the energy a fiber can withstand until it breaks, known as the breaking energy - Although this explanatory section is included within *****Industry-Specifics*****, understanding this concept is critically important for general readers in the context of environmental impact analysis. Therefore, readers who have chosen to skip this section are encouraged to review **Fig. 16**.

Variability: Prior to delving further into the analysis, it is imperative to acknowledge an inherent variability in this type of data. The characteristics of natural fibers such as cotton can exhibit significant fluctuations contingent upon various factors, including geographical origins and climatic conditions during cultivation, specific genetic variants of cotton plants, farming methods, etc. Similarly, synthetic fibers like polyester demonstrate considerable variability in their properties, largely influenced by diverse manufacturing parameters.

Consequently, various scientific literatures and other sources of information present a wide spectrum of data pertaining to the physical characteristics of these fibers. This variability underscores the importance of cautious interpretation and contextualization of the data of this nature.

Notwithstanding these variations, the dataset selected and used in **Fig. 16** represents the values typically accepted within the industry.

Strength Comparison: Notably, the polyester maintains consistent stress and strain characteristics whether dry or wet, illustrated as (3) of **Fig.16**. This is due to the hydrophobic nature of polyester and its strength is unaffected by moisture as a result. In contrast, the cotton exhibits significant differences between wet and dry, as illustrated in (1) and (2) of **Fig.16** because of its strong affinity for water, attributed to the presence of the hydroxyl groups in cellulose and the absorption of water significantly increases the cotton's strength.

The breaking energy of the polyester, represented by the red area on **Fig.16**, is significantly higher by a factor of 5 to 10 times than that of the cotton when wet and dry respectively.

In most situations other than during laundry, cotton remains relatively dry in consumption, therefore, the strength difference between the two fibers in real-world usage, which relates more closely to the creation of the dust in the air, would be closer to 10 times than 5.

This significant strength disparity can be understood with a simple analogy: **if a polyester shirt produces 1 gram of microplastics, an equivalent cotton shirt could generate nearly 10 grams of dust.**

As repeatedly mentioned earlier, variations in test results like these can arise from numerous factors and it is always necessary to take a precaution in interpreting them. However, most factors favor polyester in the aspect of the dust creation because its manufacturing variables can be easily controlled and adjusted to create more favourable outcomes, whereas cotton is subject to many uncontrollable factors like weather conditions, specific species, etc. This naturally leads to the discussion of the

next subject, where much of the variables are accounted for and result in stark differences of the dust quantities in real-world circumstances between natural and synthetic fibers.

*****Industry Specifics (2) – Start*****

Professional Experiences on Dust

During my tenure as R&D engineer in South Korea, I had an optimal opportunity to gain a profound understanding of the dust creation from different fiber types.

Quality Issues related to Dust: As a yarn manufacturer, the company I worked for occasionally received quality complaints from its customers of the weaving and knitting factories. One of the most significant and costly issues of all complaints was an excessive dust creation by some of the company's yarn products during the manufacturing processes of its customers.

Under normal circumstances, it is natural to observe some dust as yarns pass through the hard surfaces of machinery parts, often made with metals, ceramics, plastics, etc. However, if the fibers comprising the yarns are weaker than others, they produce more dust. Excessive dust becomes particularly problematic for weavers and knitters, as it accumulates and obstructs the intended passages of the yarns during their manufacturing processes. This obstruction can lead to the breakage of both yarns and small machinery parts such as needles, causing manufacturing stoppages and resulting in a material loss and waste of labor hours.

Weavers and knitters may tolerate a certain level of operational inefficiency, but they will complain when it

exceeds an acceptable level. The financial and reputational damages from such complaints prompted the company to mandate me with comprehensive studies to understand the causes of wide variations in the dust quantities in some manufacturing batches of yarns, which subsequently led to customer complaints.

Yarn Blends and Dust Quantity Analysis: The company produced yarns with a wide range of fiber compositions, including 100% Cotton, 100% Wool, 100% Polyester, 100% Acrylic, as well as various blended fibers. The blend compositions varied significantly depending on the desired properties of finished textiles, ranging from blends like 95% Wool and 5% Polyester to the reverse, and everything in between as well as widely different other fiber types mixed together. As a result, the company offered hundreds of yarn assortments to its customers.

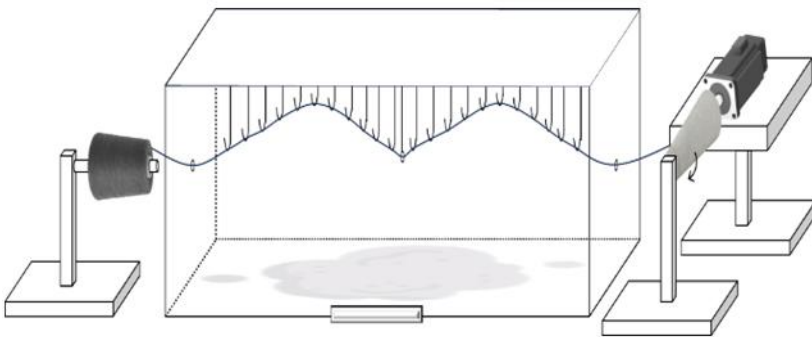
To better understand the variability in the dust generation across this wide spectrum of products, I developed a strategy to measure the nominal dust quantity for the yarns made from homogeneous materials (such as 100% Cotton, 100% Wool, 100% Polyester, etc.) and used the results as the baselines of the dust creation characteristic of each fiber type in normal quality. Then, I tried to correlate the baselines to the variation when it was blended with other fibers in varying proportions. This approach allowed me to understand how each fiber type contributed to the dust creation, regardless of specific blend ratios.

Invention of Dust Measurement Test Device: One of the challenges I had to overcome on tackling this task was accurately measuring the quantity of the dust in a controlled environment. In the open air of a typical

manufacturing facility, it was impossible to obtain precise measurements. For example, enclosing a large weaving machine in an airtight bag to collect the dust over a certain period while the machine was running would have been both impractical and extremely costly. Additionally, this method would have resulted in significant dust loss when opening and closing the bag.

To address this, I developed a compact test device that fits comfortably within the company's laboratory, as illustrated in **Fig.17**. This device featured a series of needles arranged to simulate the tensile and frictional forces subject to the yarns in conventional knitting machines. When a yarn specimen entered the chamber, it was subjected to a preset degree of physical forces before being wound on the other side. The adjustments for creating different levels of physical forces were made through the speed adjustment of the winding spool and different passage arrangements within the chamber.

Fig.17 Test Equipment for Yarn Dust Collection



In 1993, I was granted my first patent in South Korea for the device. With it, I established a nominal range of the dust quantity for varying yarn types based on different

manufacturing parameters such as fiber types and blend compositions, yarn thicknesses and twist variables, among others. If a batch of yarn produced by the company exhibited a dust generation characteristic considerably exceeding the nominal range, the batch was segregated and destroyed. This approach improved the company's operations by minimizing the customer complaints of the nature.

*****Industry Specifics (2) – End*****

Generalization of Dust Quantity: Through this experience, I established a strong correlation between the breaking energies of fibers and the amount of dust generated during subsequent manufacturing processes. For example, under identical yarn-making conditions and produced in the same pilot spinning machines, a cotton yarn created multiple times higher quantities of dust than a polyester yarn, assimilating the data presented in **Fig.16**. Although this insight could already be reasonably inferred from the comparative strength data presented in **Fig.16**, it is nonetheless essential to rigorously confirm that the same behavior persists after a fiber material has undergone complex yarn-making production processes involving numerous additional variables, starting from its fundamental scientific principles. Successfully establishing this continuity represented a significant achievement of the project.

Furthermore, this observation was applicable for all other fiber types, and it was clear that the weaker strength of natural fibers was translated directly to the higher quantity of dust created. My earlier assertion, *“if a polyester shirt produces 1 gram of microplastics, an equivalent cotton*

shirt could generate nearly 10 grams of dust”, was in part based on the conclusion obtained from this study, exemplifying the nature of scientific studies proportionally reflected in real-life situations.

Meanwhile, this conclusion, namely, the difference in the amount of dust generated from natural and synthetic fibers, has environmental implications that extend beyond a mere quantitative disparity, as these impacts are driven by both the composition and the quantity of chemical substances remaining in the particulates. This issue will be examined in greater detail in **Section 10.4**, which addresses dyeing.



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Section 4 Unfortunate Disposal

While many people may not give much thought to the environmental impacts of disposing of textile goods, there are numerous critical factors to consider. These range from the implications of the residual chemicals contained in the waste affecting our environment to humanitarian crises caused by the export of textile waste from affluent countries to underdeveloped and developing countries. In fact, the ways in which textile waste deposited in local recycling bins is handled can be deeply unsettling to those unfamiliar with the underlying processes.

4.1 Donation Myth

Many people assume that the textile goods they drop off in community recycling boxes or donate to charity groups are reused by those in need. However, the reality is starkly different from this common perception. In a 2018 episode of “Marketplace”, aired by the Canadian Broadcasting Corporation (CBC), it revealed that **only 1% of the textile waste is reclaimed and recycled** in a “real sense” of recycling. The fate of a large portion of it is mind-boggling and even exposes a troubling side of humanity.

Truth about Donation and Disposal: In other words, most of items that are ‘donated’ in affluent countries struggle to find end uses within their countries of origin. According to the most recent data released by the U.S. Environmental Protection Agency (EPA) in 2018, the average annual per-capita textile waste generated domestically is reported to reach approximately 100 pounds, or 45 kilograms.

As many people tend to assume, if this enormous volume were truly being absorbed entirely by second-hand retail stores or other recycling channels, an extraordinarily large number of such stores or exchange centers would need to exist in our immediate surroundings. In reality, however, regardless of country or city, such facilities exist only in very limited numbers, making them relatively uncommon in everyday life.

In the meantime, I had the opportunity a while ago to investigate the actual conditions of clothing donated to charitable organizations across various countries, and I was struck by two aspects in particular. First, the vast majority of donated items were in perfectly usable condition. Second, these donation-accepting organizations were, in fact, among the largest exporters of second-hand clothing in their respective countries.

Unintended Destinations: Strict environmental regulations in affluent countries make it difficult and expensive to locally incinerate or bury large quantities of textile waste. Consequently, these textiles are shipped to underdeveloped and developing countries far away from the origins of the waste. While a small fraction of them may be reused by the locals in the destinations, the majority are burned or buried without necessary environmental safeguards, such as monitoring for volatile organic compounds (VOCs) or proper analysis and control on soil and water contamination.

Fig.18 Irresponsible Textile Waste Example



<https://www.linkedin.com/pulse/tackling-ghanas-textile-waste-challenge-rémi-goget>



It is a troubling irony that thick sweaters and winter coats from wealthy northern countries end up in nations near the equator with hot weather, reflecting a typical "Not In My Back Yard" (NIMBY) attitude of the citizens of wealthy countries.

Misperceptions and Environmental Impacts: As environmental awareness has increased worldwide, many people have begun to pay closer attention to their own environmental footprint. For example, most people readily recognize that throwing plastic bottles into general waste is an environmentally undesirable action. However, those same people often give little thought to the environmental harm caused by the clothes they donate, and instead feel a sense of pride in believing they have done something good for society, creating a serious gap between their perception and reality.

This misperception carries significant environmental consequences. Beyond the carbon footprint from transporting the waste across oceans, the textiles contain residual chemicals that persist in nature, causing long-term harm, a topic explored in detail in **Chapter III: Lifecycle Assessment - Disposal** section.

Humanitarian Crisis: Perhaps the most distressing aspect of this issue is the social injustice it perpetuates. Whether intentional or not, people in affluent countries turn their blind eye on the harsh living conditions of those in the nations that receive their waste. In these regions, mountains of textile waste pile up along riverbanks or are burned in areas like school playgrounds. When viewed from even a slightly longer-term perspective, the fact that such environmental impacts can spread easily and rapidly to all regions and populations across the globe makes it difficult to describe these actions as anything other than indiscriminate and profoundly uninformed.

At this point, there is one conviction I hold firmly: if people in affluent countries were to truly understand the meaningless environmental harm and ethical implications of such practices, they would never consent to sending the goods they have used to regions such as Africa or South America. The mindset often summarized as ‘**Not In My Back Yard (NIMBY)**’ is fundamentally premised on the assumption that harm does not affect one’s own family. Once that premise collapses, it is reasonable to expect that perspectives would shift and more rational, responsible choices would follow.

4.2 Decomposition Myth

Many environmental interest groups often target synthetic fibers, due to their prolonged persistence in nature. It is commonly understood that natural fibers decompose and return to the environment much faster than synthetic fibers. However, while this basic scientific fact is true, there are many other critical factors that must be considered to fully understand the global environmental impacts of natural versus synthetic fibers.

Residual Chemicals: Although natural fibers like cotton (cellulose-based) and wool (protein-based) degrade more quickly in nature, the chemicals used during their manufacturing and those left in the textile waste present a significant challenge in the environmental management. For instance, various colors in the textile products indicate the presence of the dye pigments and numerous other processing chemicals. These substances can leach into soil, contaminating runoff and groundwater in landfills, thereby entering the ecosystem. When incinerated, they pollute air and can spread globally through the atmospheric circulation.

While more detailed information will be provided in **Chapter II: Basics** and **Chapter III: Lifecycle Assessment**, it is important to recognize that natural fibers often require significantly larger amounts of harmful chemicals during production compared to synthetic fibers, primarily due to their heightened chemical reactivity. Furthermore, the chemicals used for natural fibers are typically more reactive, as they are formulated to interact with the chemically active functional groups in these fibers, whereas the chemicals

used for synthetic fibers generally do not depend on chemical reactions in the same manner.

Thus, understanding the quantity and toxicity of the chemicals involved in both natural and synthetic fibers is crucial for accurately assessing their true environmental impacts.

Environmental Impacts

Toxicity of Pesticides for Farming Natural Fibers: Natural fibers, particularly the plant-based ones like cotton, are exposed to toxic chemicals from the very beginning of their lifecycles. For example, pesticides used in farming are designed to eliminate living organisms like insects and bugs and are therefore highly toxic. In high concentrations, they can even significantly harm humans and large animals. In fact, the toxicity of these chemicals far exceeds that used in most other textile manufacturing processes such as dyeing. Much of this toxicity remains in the environment or in the textile materials themselves for periods that can far exceed multiples of human lifetime.

Dye Types Overview: Among all the stages of textile production outside the cultivation of natural fibers, the dyeing is typically most chemically intensive and creates the highest environmental impacts. Different types of fibers require different dyes and associated chemicals based on their chemical compositions. For example, cotton and wool require highly reactive dyes, known as "Reactive Dyes", while polyester uses a different type, known as "Disperse Dyes". As the name indicates, the reactive dyes contain chemical elements and substances which promote reactions with the chemical groups of the targeted fibers

whereas the disperse dyes are designed to physically impregnate into the polymer structures of synthetic fibers.

Readers familiar with hair dyeing and washing plastic food containers can easily draw parallels to textile dyeing. Like wool dyes, which are reactive and designed to chemically bond with the protein in wool, the hair dyes also involve a chemical reaction with the protein contained in the human hair. The dyed hair color fades over time relatively quickly, and proportionally with the frequency of washing. This can be compared with the colors on plastics, often found in food container lids, etc. These colors never fade regardless of how many times they are washed because they do not interact with water and does not shed color pigments. More detailed information on the dyeing related aspects will be provided in the following discussions including the dedicated **Chapter II: 10.4.2 Dye Types and Toxicities.**

Chemical Reactivity and Environmental Impacts: In general, the environmental impacts of the reactive chemicals used in textile production relate to both their chemical reactivity and their persistence in the environment, either in their original forms or as the byproducts of associated chemical reactions. These impacts are particularly significant when more reactive substances, such as salts, heavy metals, etc., are involved in the process. On the other hand, less- or non-reactive chemicals, like the disperse dyes, tend to have environmental impacts more due to their persistence on their own rather than chemically reacting with their surroundings or creating byproducts.

Natural Existence and Densification: Although many of the chemical elements and compounds used in the textile

manufacturing exist naturally in the environment, their concentrations are typically low and do not pose health risks to humans and animals under usual circumstances. However, when these are artificially concentrated to achieve specific characteristics, their toxicity increases, posing significant threats to the living organisms and environmental sustainability. Pesticides, fertilizers, dyes, and various other chemicals used in the textiles are examples of these concentrated substances. Some of these compounds are known to cause severe health effects, including cancer, coma, and seizures.

Different chemical treatments required for natural versus synthetic fibers, while both are harmful, can lead to significantly different environmental outcomes. This topic is explored in detail throughout **Chapter II** and **Chapter III**.

4.2.1 Duration of Decomposition

There is a myriad of information online regarding the decomposition durations of synthetic materials, but much of it lacks scientific validity, leading to widespread public misconceptions and associated environmental harm. An example is illustrated in **Fig.19**, where it is claimed that plastic bottles and diapers last for 450 years in nature. Other sources suggest even longer durations, some stating that plastic bottles may persist for 650 years or even indefinitely. These claims, often ranking high in various search engines like Google, contribute to the misinformation crisis we face today – more discussions on this to be presented in **Chapter IV: Misinformation Crisis**.

Fig.19 Example of Plastic Duration in Nature on Internet



Validity of Information

Plastic was invented in 1907 and has been around for just over a century. Naturally, scientific and analytical minds might question the claims about plastic persisting in nature for hundreds of years. Some might argue, *"If the modern science can calculate the Earth's age at 4.55 billion years*, estimating plastic's longevity in the range of hundreds of years shouldn't be difficult."* However, this line of thinking overlooks the fundamental principles behind the underlying science.

* The figure established by Dr. Clair Patterson and widely accepted in the scientific community

Radioactive vs. Chemically Inert Materials: The methods used to determine the age of Earth and also ancient artifacts rely on the radioactive decay of certain elements. For example, Dr. Patterson used the half-life of a specific

lead isotope to calculate the Earth's age where the half-life refers to the time it takes for half of the atoms in a radioactive substance to decay.

In other words, for such a calculation method to be valid, the material in question must contain chemical elements that are suitable for isotopic age dating. This approach relies on the properties of radioactive isotopes, such as carbon-14 (^{14}C) present in once-living organisms, or other radioactive isotopes found in certain inorganic materials. Plastics, by contrast, are chemically inert materials and do not contain any elements that allow for isotopic age dating. Consequently, it is fundamentally impossible to estimate the degradation timescale of plastics using the same scientific principles employed to determine the age of the Earth.

While there are scientific methods for studying the plastic degradation under specific conditions, these methods are based on vastly different scientific principles and cannot be used to accurately predict plastic's decomposition timeline in the context of several hundreds of years. While related scientific details will be discussed in subsequent discussions, it is clear that widely circulated information on plastic decomposition such as the earlier “450 years” example lacks a solid scientific basis.

Plastic Decomposition and Evidence

As such, one common misunderstanding in the textile sustainability is the perceived longevity of synthetic fibers (or plastics in general) in nature. To better understand this, let's travel back in time to Seoul, South Korea, between 1978 and 1993. During this period, Nanji-do served as a

massive landfill for the greater Seoul area, receiving an average of 3,000 truckloads of household waste daily. By the time it closed, it was the tallest landfill in the world, containing the total of 93 million tons of waste.

Personal Experience: I have a personal anecdote which relates to this site of Nanji-do. In 1995, as a young professional freshly landed in Seoul, I decided to save money on driving lessons by practicing at Nanji-do. Unlicensed driving instructors used the landfill site to draw parking and practice routes on flattened areas on top of the waste and offered lessons at much cheaper rates than licensed ones in the city. I commuted there for the lessons, enduring severe odor coming from the decaying waste and observed firsthand the compositions of the waste, including large amounts of plastics (see **Fig.20** for the examples of colorful plastic waste).

Fig.20 Nanji-do in the 90's



After I immigrated to Canada in the late 1990s, Nanji-do faded from my memory. Then, years later, I came across a news article reporting that the landfill had been transformed into a part of the Seoul Sangam World Cup

Stadium Park. Interestingly, a portion of Nanji-do was designated for a power plant that generates electricity using the gas from the decomposition of the household waste, including plastics. From the landfill's closure to the extraction of this gas, it took about 30 years.

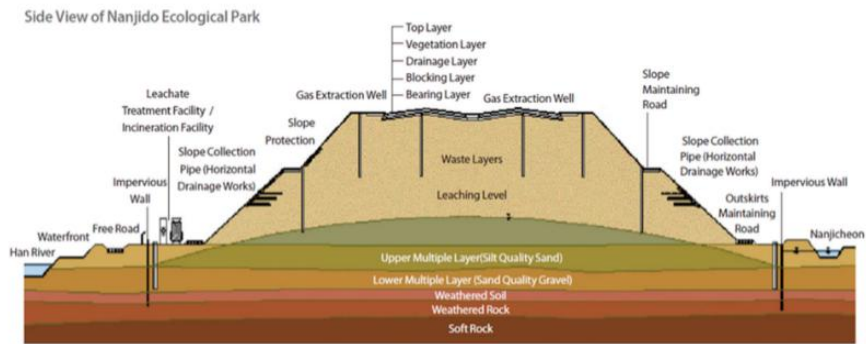
Fig.21 Nanji-do NOW



Image: Seoul WorldCup Stadium Park



Image: Power Plant



Our Own Observations: Now, let's fly out of Seoul and return to our own backyards and surroundings. Many of us have encountered plastic pieces left in the environment, perhaps picked up while walking or gardening. If you have purchased plants from a local store, you may have noticed that they often come wrapped in black plastic bags around the soil.

Significant Strength Degradation: Our experiences with unused or freshly used plastic bags give us a baseline for the typical strength of plastics in daily use. Compare this to the plastics used to wrap the soil, which often show significant strength degradation after being in contact with the soil for a relatively short time. In the perspective of the plastic manufacturing, it is implausible to intentionally make the initial quality of the plastics so fragile due to their inherent characteristics related to the polymerization principles. Although not a perfect "apples-to-apples" comparison, we can easily estimate that these plastics weakened over time as they tear or puncture far more easily than the baseline reference based on our experiences, suggesting that the decrease in the strength was associated with the soil contact.

Science of Degradation: In scientific terms, the degradation of the plastic strength indicates that the polymer chains inside the plastics broke down. To illustrate this concept, imagine a fishnet: if a few of its connecting parts are broken, the net's overall structure becomes compromised and the strength decreases. Similarly, when the bonds within the plastic's polymer chains break, the material loses its strength.

Practical Experiment: A curious mind can easily verify this phenomenon at home: Bury a piece of plastic bag under wet soil in a shaded area and leave it for a month or two while keeping another identical piece in a kitchen cabinet. After a certain duration, the buried plastic will have significantly lost its strength, while the stored plastic will remain intact. This experiment is a practical way to observe the plastic degradation without having to wait "hundreds of years", the

durations often cited on the internet - I recommended using plastic bags used in grocery shopping for this experiment because the effects will be more readily noticeable in a relatively short time frame and can be verified without the need of laboratory equipment.

In summary, most plastics are chemically inert and do not contain elements that can be used for isotopic dating methods. As a result, the degradation processes of plastics are governed by scientific principles that are entirely different from those used to determine the age of ancient artifacts. The degradation is, in fact, due to the existence of the plastic-eating bacteria in nature. In certain conditions, such as chemically active landfills, these bacteria thrive in higher densities and exhibit increased level of activities. Certain chemical compositions, temperatures, and pressures present in such landfills significantly aid these bacteria in degrading plastics at a much faster rate than what many sources suggest on the internet. The Nanji-do landfill is a prime example of this accelerated degradation process. Furthermore, the subsequent discussion of creating the artificial conditions of plastic decomposition is another example.

Artificial Conditions of Decomposition: Another piece of evidence comes from various “*biodegradability*” claims made by many companies today on certain plastics. In some laboratory or manufacturing environments, specific conditions such as particular bacterial compositions, temperature, pressure, etc., can be created for plastic degradation, the conditions reflective of actual landfills

(anaerobic degradation), or left in the open air (aerobic degradation).

While humans can create and adjust these conditions for accelerating or decelerating the degradation rates, they do not create the bacteria themselves. These organisms naturally exist in the environment and constantly at work to break down the materials left in nature. Fortunately, plastics also serve as a viable food source for these microorganisms. By definition, therefore, all plastics are biodegradable and many “*biodegradability*” claims, unfortunately, are simply the exploitation of the public interests on the environmental sustainability with the misrepresentation of the underlying science for their benefits.

Variabilities of Plastic Decomposition: Different pieces of plastics degrade at varying speeds. The form factor of plastics whether they are thin and light like plastic bags and bottles or thicker and heavier like food containers affects how easily the bacteria can access and break down the polymer chains. Additionally, the density of plastics plays a crucial role: the higher the density, the more challenging it is for the bacteria to penetrate the polymer structures and break the links.

4.3 Biodegradability

The term “*biodegradability*” has recently become a major source of misinformation and greenwashing by many organizations, exploiting the public’s keen interest and pursuit of the environmental sustainability. In this subsection, I will debunk some myths surrounding “*biodegradable*” plastics.

Before diving in, it is important for the public to understand that the term "*biodegradable*" by itself does not necessarily indicate an environmental benefit. There are many factors involved in this simple term, and without understanding these nuances, the public can be misled as the term is often used out of context.

Definitions

The first step is to clearly define the "*biodegradability*". A plastic is considered biodegradable if it breaks down under natural conditions. As discussed earlier, however, all plastics are technically biodegradable due to the activities of the plastic-eating bacteria in nature. Given enough time, all plastics will eventually degrade. Therefore, this discussion begins with highlighting the inherent flaw in "*biodegradable*" claims.

Context: In the context of labeling a plastic as "*biodegradable*", specific conditions, typically in laboratory settings, are used to assess its degradation characteristics in a certain form factor under controlled environment. These conditions involve carefully regulated factors such as bacterial activity, temperature, pressure, and chemical composition in relation with a time factor.

However, this approach has limitations. For example, a thin plastic might be classified as "*biodegradable*" based on the test results obtained under certain conditions, while a thicker piece of the same polymer may not be, due to the differences in how easily bacteria can access the polymer chains and the rate of degradation. Additionally, real-world conditions in landfills vary significantly across the globe, making it unrealistic to expect that a plastic labeled

"*biodegradable*" will degrade the same way in diverse environmental conditions as it did in the controlled test environment.

Because of these limitations, the term "*biodegradable*" can be misused by profit-seeking organizations as a tool to create misleading perceptions among consumers. Moreover, in analyzing such information, one critical point that must not be overlooked is that plastics lacking a "*biodegradable*" label cannot automatically be assumed to be non-degradable in natural environments.

Consequently, the following discussions will involve an analysis that may appear ambiguous or somewhat confusing, as they address methodologies in which materials that are fundamentally and intrinsically biodegradable are classified and marketed as "*biodegradable*" in ways that can mislead consumers. For clarity, readers are asked to note that, among the various uses of the term "*biodegradable*" that follow, those areas in which misunderstanding may arise due to common corporate practices are presented in quotation marks and italicized.

Bioplastic: The term "*bioplastic*" refers to the plastics that are produced from natural biomass sources of either plants or animals such as corn starch, woodchips, proteins, etc., or the presence of bacteria in their synthesis. However, not all bioplastics can be identified as "*biodegradable*" according to the definition mentioned earlier. For example, bio-based PET (polyethylene terephthalate) is a non-"*biodegradable*" bioplastic. Even though it is synthesized with the presence of bacteria, the chemical structure of the bio-based PET remains identical to its fossil fuel-derived

counterparts, including its decomposition properties. Some studies show that certain bioplastics can be manufactured with a lower carbon footprint than their fossil counterparts while other bioplastics' processes are less efficient and result in a higher carbon footprint than fossil plastics.

Although more comprehensive understanding is required rather than simply accepting the results of these studies, it can be easily concluded that we cannot automatically assume an environmental benefit from the term of 'bioplastic', the theme which applies for the remainder of the discussions of this subject.

Oxo-Degradable Plastics: Another example is 'oxo-degradable' plastics. Despite their seemingly eco-friendly name, they are essentially conventional plastics with additives known as pro-degradants that accelerate oxidation process under specific conditions involving oxygen. While oxo-degradable plastics break down more quickly when exposed to sunlight and oxygen, conditions typically found in open air rather than in landfills or water, this process merely fragments the plastic into microplastics. As a result, the perceived benefit can lead to other environmental and health risks as microplastics contaminate air and water, ultimately entering the food chain and posing threats to both human and animal health.

Compostable Plastics: In some cases, compostable plastics are labeled as "*biodegradable*". Compostable plastics decompose within a specified timeframe under controlled conditions typically found in industrial composting facilities. These facilities create optimal environments for decomposition, including high

temperatures, controlled pressure, and specific chemical conditions. Accelerating the decomposition rate of otherwise-naturally-degrading plastics renders this factory-based decomposition process economically inviable. Thus, there are very few such facilities, and they are generally used for specialized purposes.

Degradability of Plastics: An important aspect of these discussions is the natural degradability of all plastics, as repeatedly emphasized throughout this section. To substantiate claims of more favorable environmental impacts, a “*biodegradable*” plastic must be proven to degrade in natural conditions, such as in landfills, air, or water, considerably faster than other plastics, and without creating harmful side effects.

With this premise in mind, it is beneficial to review the scopes of a typical test method used to label plastics as “*biodegradable*” and other related claims.

*****Industry Specifics (3) – Start*****

ASTM D5511 – 18

Excerpts of the test method includes the following statements:

This test method is designed to yield a percentage of conversion of carbon in the sample to carbon in the gaseous form under conditions found in high-solids anaerobic digesters, treating municipal solid waste. This test method may also resemble some conditions in biologically active landfills where the gas generated is recovered and biogas production is actively promoted by inoculation (for example, code position of anaerobic

sewage sludge, anaerobic leachate recirculation), moisture control (for example, leachate recirculation), and temperature control (for example, short-term injection of oxygen, heating of recirculated leachate).

Claims of performance shall be limited to the numerical result obtained in the test and not be used for unqualified “biodegradable” claims. Reports shall clearly state the percentage of net gaseous carbon generation for both the test and reference samples at the completion of the test. Furthermore, results shall not be extrapolated past the actual duration of the test.

*****Industry Specifics (3) – End*****

As a result, the outcomes of this test method **can only be presented as numeric data and cannot be used to justify “biodegradable” claims**. Several other test methods under similar scope exist globally.

Corporate Misconduct: Unfortunately, many companies exploit the fact that consumers are unlikely to fully understand the implications of such claims, taking advantage of this loophole. While this approach is clearly unethical, such corporate misconduct is not uncommon. Companies often present limited information without proper context, leading consumers to instant misperceptions. This behavior is typical of the organizations that prioritize short-term gains over transparency, due diligence and honest effort to truly benefit the environment.

Even if we extend the benefit of doubt and assume that these companies lacked the necessary knowledge and acted unknowingly, this does not absolve them of their social and environmental responsibilities. A couple of hours

of thorough reading would be enough to grasp the basics of the subject matters and guide them to avoid being engaged in such acts. Failing to put in even this minimal effort and commitment can only be described as misconduct and greenwashing. Having spent decades in the global textile industry, I find it deplorable to witness this level of dishonesty even from some of the most reputable brands which undeservedly enjoy a high level of consumer trust.

Poor Business Practices and Greenwashing

Based on the foregoing discussion, companies that claim their products to be “*biodegradable*” should not limit themselves to merely affixing such a label. Instead, they must clearly and transparently disclose all relevant information so that consumers can readily understand the scope and limitations of the testing methods used. In practice, many countries, including the United States, the European Union, Australia, and Japan, have established regulations aimed at preventing companies from using the term “*biodegradable*” as a tool for greenwashing. Nevertheless, the lack of effective oversight and enforcement by governments in many countries continues to allow numerous companies to advance such claims without adequate evidence or sufficient explanation, thereby misleading consumers.

“*Biodegradable*” Thermal Insulation?: Some companies exploit other loopholes in “*biodegradability*”. For example, a company in the thermal insulation industry claims that one of its products is “*biodegradable*”.

In manufacturing, thermal insulation materials are sandwiched between lining fabrics and enclosed within the

structure of finished goods. Among these thermal insulation materials, those made from synthetic fibers are also commonly used due to their various advantages, and the company in question claims that its synthetic fiber-based material is “*biodegradable*”. However, a scientific examination of this claim reveals that it gives rise to two major contradictions.

First, the structural configuration of thermally insulated products is such that the insulating materials are enclosed by outer fabrics with highly diverse properties, preventing them from being directly exposed to the natural environment for extended periods of time even in burial. These outer fabrics vary widely depending on the intended application, ranging from lightweight jackets to thick, heavy winter outerwear, ski apparel, and workwear, and are selected according to the final product’s appearance and functional requirements. As a result, the materials used can differ dramatically in their properties: some are thin and lightweight, while others are thick, heavy, or coated with polymer-based membranes. All of these outer material types act as barriers that delay microbial access required for biodegradation. Consequently, the diversity of enclosing fabrics implies that the rate at which microorganisms can access the insulating material does not follow a consistent pattern but instead varies significantly depending on the specific outer fabric used.

Second, when testing “*biodegradable*” materials, they are evaluated in isolation, without considering other components of finished products. Therefore, this approach of using the “*biodegradable*” conclusion obtained from such method works for single-layer items, such as shirts or

pants, where the fabrics in question can be directly exposed to similar decomposition conditions in nature.

However, for multi-layered structures like winter jackets or heavy blankets that contain thermal insulation materials inside, any “*biodegradability*” claim about the insulation material must also account for various lining fabrics used.

In summary, given the wide variability in the fabrics and structural configurations of finished products, it is virtually impossible to reliably verify the “*biodegradability*” of thermal insulation materials under natural conditions. The sheer number of potential combinations of product structures and outer fabric types precludes comprehensive testing and validation sufficient to support such a claim.

Just Another Exploitation Example: Yet, the thermal insulation company continues to make their unsubstantiated “*biodegradability*” claims without addressing these concerns. Then, manufacturers and brands accept such claims and transfer them to consumers via various marketing media and tools attached on the products they sell. As previously discussed, the public's lack of knowledge enables the prevalence of numerous cases of exploitation by these greenwashing organizations.

What I find particularly regrettable is that this issue extends beyond the perspective of intermediate material suppliers who wish to market their insulation materials with environmentally friendly labels for unjustified reasons. Even among the many individuals working for manufacturers or brands that produce and sell finished products, it is difficult to understand why such claims, requiring only the most basic level of knowledge and analytical reasoning to challenge, are left unexamined and transmitted to

consumers without scrutiny. I believe that all readers of this book will be able to share this sense of disappointment. Moreover, regardless of whether such unethical practices arise from a lack of capability or a lack of effort on the part of these companies, it is easy to recognize that they ultimately amount to what can only be described as greenwashing. Finally, it is deeply troubling that such examples occur so widely and so frequently around us that cases of this nature can be presented repeatedly throughout nearly the entire book.

As a consumer, I do not consider accelerated biodegradability of plastics to be beneficial to my lifestyle. If a plastic's biodegradability is significantly enhanced compared to regular plastics, I would be concerned about its degradation during use. I keep most of my textile goods for decades, such as the very first suit I wore for my university graduations, still hanging in my closet and occasionally worn. I would not want them to degrade in my closet and continuous use.

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Section 5 Unfortunate Recycling

Many people form opinions and perspectives on various aspects of life based on single-word descriptions or short phrases. "Recycling" is one such example. Given the global importance of the environmental sustainability, one might expect questions like, *"Is recycling done without creating negative environmental impacts?"* to be common. However, this is often not the case.

Face value: Even among the most educated members of our society, there is a tendency to accept convenient notions and purchase textiles with such brief sustainability claims. Labels such as "Recycled" or *"Made from x number of plastic bottles"* are often taken at face value. This point is readily demonstrated by the way in which such simplistic and seemingly clear-cut claims are accepted at large and come to shape the underlying consciousness of our society. In this sense, it can even be argued that these opinion-leading groups effectively position themselves as supporters and enablers of greenwashing.

Serious Gap: There are several key reasons for this phenomenon. One is that consumers often place too much trust in the companies making various sustainability claims. Furthermore, consumers may not be trained to ask critical questions and often lack diligence to seek truth on more complex issues which do not create more immediate and observable impacts in their lives. Unfortunately, environmental matters are of such characteristics. They readily and uncritically accept values and claims that are neither directly visible nor immediately perceptible in their own lived reality, doing so from a stance of complacent

ignorance. Together, these factors create a significant gap that allows for numerous corporate misconducts, where untrained consumers are exploited.

Given the prevalence of greenwashing, even by some of the most trusted organizations globally, the public's overly simplistic approach to various sustainability claims does little to benefit the environment. It is crucial for consumers to take a more critical stance and demand transparent, comprehensive information in order to make informed decisions that truly benefit the environment – Quote of the Chapter I.

Recycling Myth

Plastic recycling can be conducted in various ways, but the most conventional methods unfortunately result in critically negative environmental impacts. In fact, we do not even know if there is a clear net benefit from the current global recycling efforts. For many readers, this statement may be surprising, as most consumers believe that purchasing goods made from recycled contents is beneficial for the environment. Here, I present the information that may challenge this common understanding.

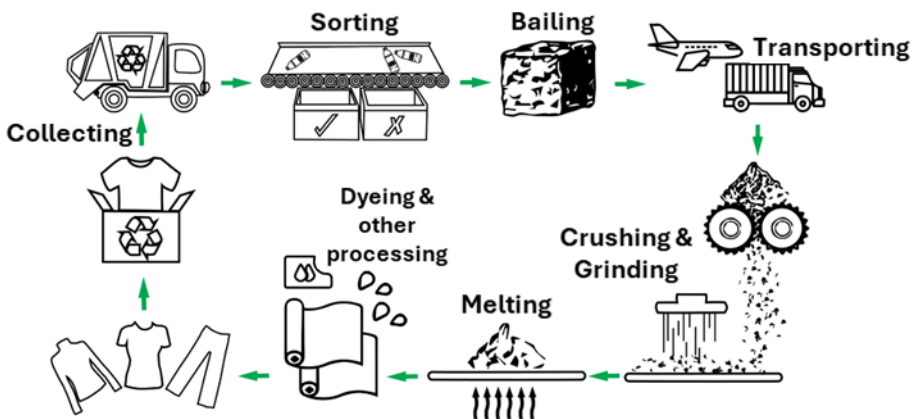
Importation Ban on Plastic Waste: Many people may recall the garbage crisis that most Western countries experienced in the late 2010s, which was precipitated by China's total ban on the importation of plastic waste, the primary source for plastic recycling - For years, advanced economies had been exporting large quantities of their plastic waste to China and other countries, and this sudden ban led to local waste crises.

China's Gradual Actions and Similar Actions by Other Countries:

Although the ban seemed abrupt to the Western world, the Chinese government had initiated related measures as early as 2013 with the CHINA GREEN FENCE policy. This was followed by the CHINA NATIONAL SWORD policy in 2017, culminating in the total ban in 2018. Subsequently, many other countries, such as Vietnam, Malaysia, Thailand, etc., implemented similar measures, focusing on processing only their own domestic plastic waste. The actions taken by these governments stem from a clear understanding of the negative environmental impacts associated with the conventional recycling processes and are now spread to a large number of other underdeveloped and developing countries around the world.

In this discussion, I will elucidate four main areas of negative impacts caused by the conventional methods, accompanied by the flow map depicted in **Fig.22**.

Fig.22 Overview of Conventional Plastic Recycling



Inefficient Carbon Footprint: The processes result in a significantly inefficient carbon footprint due to the transportation of plastic waste across international oceans,

as most recycling facilities are located in countries far from the consumption centers where the waste is generated. Given that a substantial portion of the plastics used in the products and packaging originates from these distant countries to begin with, shipping the waste back to the regions of origin exacerbates the inefficiency in the carbon footprint management

Microplastics in Manufacturing: Most conventional recycling is carried out through a process known as “crushing” and “grinding”, shown by “2” in **Fig.22**. This method involves applying substantial physical force to break the waste into small pieces, which are then submerged in a large pool of water.

The purpose of this process is to remove dirt and foreign materials contained in the waste. Observations at the storage yards of recycling facilities readily reveal such heavy contamination present in the plastic waste, waiting to be processed.

Fig.23 Contaminated Plastic Bottles



Given the diverse shapes, sizes, and types of contamination present, the only commercially viable way is to crush the plastics and break them into small pieces, then use water

to wash away the contaminants. **Fig.24** illustrates the stage of the process where the crushed plastics are deposited into a large quantity of water.

Fig.24 Crushed Plastics



The application of heavy physical forces to break down the plastics inevitably generates a significant amount of debris, including microplastics.

Strength Factor and Environmental Harm: Despite the extensive cleaning efforts, which involve heavy water and electricity usage, microplastic generation, and potential chemical use to dissolve certain types of contaminants, it is virtually impossible to remove all foreign materials. As a result, significant quantities of these contaminants persist throughout the subsequent recycling processes and become integrated into the recycled fibers.

These contaminated recycled plastics exhibit distinctive colors. Higher proportions of contaminants result in a more greyish tone in the polymer pellets, as shown in **Fig.25**,

which are then converted into fibers, carrying the grey tone over. This color difference cannot be detected by consumers, as various efforts are made during subsequent dyeing processes to conceal such imperfections.

Fig.25 Recycled Pellets (left) vs. Virgin Pellets (right)



The presence of contaminants in the recycled fibers adversely affects their strength as it hinders the formation of polymer structures during the subsequent melting process. As a result, most recycled fibers are invariably much weaker than their virgin counterparts. In some cases, the tensile stress and strain, and consequently the total breaking energy, of recycled fibers are less than 50% of those of virgin fibers.

This phenomenon occurs because contaminants interfere with the firm fixation of pigments and various chemical substances within the polymer structure. When this is considered in terms of environmental impacts in everyday life, it can be understood that recycled fibers with approximately 50% of the strength of virgin fibers generate roughly 50% more particulate matter, along with the chemical substances contained within it.

Harmful Chemicals: Recycled fibers undergo identical manufacturing and consumption processes as virgin fibers,

including chemically intensive dyeing process. It is well-known within the industry that different dyeing formulations must be used for recycled fibers to compensate the varying degrees of contamination to achieve desired color tones and hues whereas pure virgin fibers do not present such issue. This necessity may lead to increased usage of color pigments and other chemicals to counteract their inherent greyish tones.

In conclusion, there is no clear evidence to suggest that the conventional recycling efforts yield a net benefit in the overall environmental management. In my view, the disadvantages can easily outweigh the benefits. Increased quantities of microplastics alone could justify ceasing these activities altogether.

I have long argued that, given today's inefficiencies in plastic recycling, resulting in considerable harm, the best option is to place plastics in landfills. Unlike textile waste, plastics generally do not contain harmful dyes or chemicals. Even colorized plastics likely contain non-reactive dyes that do not interact with the environment. While plastics may remain in landfills for extended periods, they are less likely to contribute to other significant environmental harms.

The good news is that we do have a viable solution based on the principles of successful plastic recycling. This topic will be further explored in **Chapter V: Solutions**, specifically in the **Clean Recycling Initiative™** section.

Chapter I: Post-Chapter Commentary

In this chapter, I discussed various topics related to consumers and the textile products they consume, along with the environmental issues involved. I made a concerted effort to present the relevant science in simple terms to ensure all readers, regardless of their prior knowledge or experience, can transition smoothly into gaining a more fundamental understanding of textile materials and science in **Chapter II: Basics**.

Along with many examples shared in this chapter, readers will gain deeper insights in the next chapter into the widespread misunderstandings and misconceptions that arise from oversimplified ideas, such as the flawed concept of "*fast fashion*" and the unsubstantiated claims that plastic bottles take 450 years to decompose. Unfortunately, these misconceptions dominate public perception and shape society's collective mindset about what is good or bad for the environment, often inadvertently undermining global sustainability efforts.

Consumers, Be Aware and Beware!: If we continue to allow these mistakes and mismanagement under the guise of environmental sustainability, we may reach a point of no return. In my view, spreading accurate, science-based information is the first step to take to mitigate the negative impacts from such undesirable actions.

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What Awaits You in the Rest of the Journey

Chapter 2, 3, 4 and 5 are not included in this distributable edition.

If the ideas in this chapter have challenged your assumptions or sparked new questions, the journey is only beginning. What you have read so far exposes the symptoms of a much larger issue. The following chapters uncover the scientific foundations, analytical tools, and systemic solutions necessary to navigate environmental sustainability with clarity and confidence.

Environmental sustainability cannot be built on intuition, popularity, or good intentions alone. Without a solid grounding in science, sustainability becomes guesswork, and guesswork at a global scale can lead to consequences far greater than the problems we aim to solve. To distinguish meaningful solutions from misleading narratives, it is critical to develop disciplined, science-based understanding. The remaining chapters provide that foundation.

See brief summaries of the rest of the book on the following page.

Chapter II – The Science Beneath the Surface

If Chapter I challenges what we think we know, **Chapter II builds the scientific foundation** needed to replace misinformation with clarity. This chapter explores the chemistry and engineering behind natural, synthetic, and semi-synthetic fibers, as well as the manufacturing processes that determine their performance and environmental impacts.

By understanding how cotton, polyester, wool, and other fibers are actually produced and processed, readers move beyond emotional narratives and surface-level claims. Sustainability is not decided by labels or trends. Rather, it is governed by chemistry, physics, and industrial realities. This chapter equips you with the scientific literacy necessary to evaluate materials independently.

Chapter III – Lifecycle Assessment: Seeing the Whole Picture

Environmental debates often isolate a single factor, usually carbon, while ignoring broader consequences. **Chapter III introduces a comprehensive Lifecycle Assessment (LCA)** framework, examining impacts from farming or polymer production all the way through consumption and disposal.

Using a sixteen-point evaluation system, this chapter reveals how partial analyses can lead to deeply misleading conclusions. Readers discover that what appears “green” in one stage may create greater harm in another. This chapter strengthens holistic thinking - a discipline essential not only in textiles, but in all sustainability decisions.

Chapter IV – Greenwashing and the Misinformation Crisis

Chapter IV confronts one of the defining problems of our era: **the widespread distortion of environmental information.** Well-meaning consumers, advocacy groups, corporations, and even institutions often operate within incomplete or flawed frameworks.

Through case studies and real-world examples, this chapter exposes certification loopholes, oversimplified claims, and systemic weaknesses in how sustainability is communicated. Becoming a Green Hero requires intellectual discipline - the courage to question popular narratives and demand comprehensive evidence.

Chapter V – Solutions and United Efforts

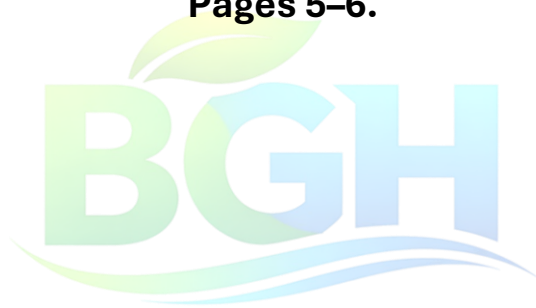
Identifying problems is only the first step. **Chapter V shifts toward constructive pathways forward.**

It presents practical approaches grounded in science-based education, improved transparency, collaborative governance, and technological innovation, including the **Clean Recycling Initiative™** and the broader **BE GREEN HEROES (BGH)** movement.

This chapter emphasizes that meaningful environmental progress cannot rely on isolated efforts or emotional campaigns. It requires informed citizens, responsible industries, and coordinated global action guided by knowledge rather than ideology.

The journey toward environmental literacy does not end with awareness. It begins there. Continue the full exploration and deepen your understanding by reading the complete *BE GREEN HEROES* book.

To access the complete text of *BE GREEN HEROES*, please refer to the links provided on Pages 5–6.



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Closing Thoughts

Fear Factor

Our lives are filled with diverse emotions, from positive feelings like joy and hope to negative ones like fear and guilt. Unfortunately, negative emotions often have a more profound impact on us. While feelings of overjoy don't usually require medical attention, anxiety might lead us to seek treatment. This explains why bad news travels faster and engages us more deeply, particularly through the fear of unknown futures.

Nature of Predictions: This phenomenon is evident in the collective social mindset that shapes our behaviors and reactions to predictions. Often, such predictions provoke fear and deeply influence our perceptions. A historical example is the widespread interpretation of Nostradamus's cryptic prophecies as foretelling the world's end at the close of the last millennium, which became common public knowledge for centuries and stirred significant concern for many.

More directly related to the theme of this book, I remember the predictions made during the 1970s and 1980s that warned of imminent oil shortages. These forecasts suggested that oil reserves would be depleted within 30 years, implying that by the early 2010s, the world would face an energy crisis that could plunge us back into a primitive way of life. These dire predictions were put forth by individuals considered experts at the time, such as industry engineers, PhDs, professors, and seasoned journalists who presented seemingly reliable scientific data such as current

reserve levels and rising consumption rates to support their claims of looming depletion.

However, this disastrous outcome has not occurred. In fact, today we have more oil reserves than we did in the 1970s and 1980s, thanks to advanced technologies that have discovered new deposits. For example, the recent shale gas revolution, utilizing hydraulic fracturing and horizontal drilling, has enabled extraction in areas previously not considered as reserves. As a result, many experts now estimate that we have more oil reserves than we will ever need as the world rapidly shifts towards renewable and nuclear energy sources.

Nature of Threats: It goes without saying that we must do everything in our power to address the challenges of climate change. The scope of its consequences can be overwhelming, and while there are countless concerns, individuals tend to have varying degrees of concerns on different issues. Some are more alarmed by certain phenomena, while others are more influenced by different aspects. This is only natural, given the vast number of environmental problems we face today. When it comes to carbon impacts, my greatest concern lies with the phenomena occurring in formerly permafrost regions, where methane (CH₄) gases are erupting from the ground, bubbling up in lakes, and ancient pathogens are being revived.

Among the key perspectives I present in this book, the central message is that our sustainability efforts must not become trapped in narrow viewpoints. In other words, it is critically important not to overlook environmental pollutants with inherent toxicity, whose cumulative effects

can produce more immediate and severe consequences, simply because attention is overwhelmingly focused on carbon emissions at the moment.

The science is clear: cumulative effects from both reactive and inert chemicals dictate whether the consequences are more immediate or gradual. While climate change has taken the center stage of global environmental efforts, the broader implications of these chemically reactive substances must not be overlooked.

Overcoming Challenges

Although I have addressed many serious concerns throughout this book, the reality is that the world is not coming to end any time soon whether by the causes and views expressed by some climate extremists or the toxicity concerns I have raised.

Reflecting on the history of Earth and humanity, we understand that the most likely causes of catastrophic events remain large asteroid or comet strikes, massive volcanic eruptions, or devastating microbial outbreaks. From a climate perspective, science reveals that cooling, rather than warming, has historically caused greater damage. For instance, the Ordovician-Silurian Extinction (~444 million years ago) saw the disappearance of approximately 85% of marine species, largely due to global cooling.

I acknowledge that this statement might spark debate, especially in the context of ongoing discussions about climate change. However, this book does not aim to take sides in the debate over the validity or extent of the ongoing climate change's impacts. Instead, I encourage readers to

approach the topic with a balanced perspective, as there is often more to the story than what is presented. While climate activism dominates public discourse, it is essential to consider insights from different viewpoints. With that in mind, I recommend two books that offer important perspectives on environmental efforts:

* ***Not the End of the World: How We Can Be the First Generation to Build a Sustainable Planet*** by Hannah Ritchie

* ***False Alarm: How Climate Change Panic Costs Us Trillions, Hurts the Poor, and Fails to Fix the Planet*** by Bjorn Lomborg

Naturally, there are many parts of the books I agree with and others I do not. However, an important message of these books, which I also strive to convey herein, is that we must avoid panic. We are not doomed.

The Dutch have successfully managed rising sea levels since the 14th century, and humanity has endured changing weather conditions since prehistoric times. Early humans, such as Homo Erectus, survived multiple ice ages without modern luxuries like insulated homes or advanced winter gear. Humanity's resilience and adaptability have allowed us to thrive despite numerous climate challenges.

This adaptability extends also to the way we address current environmental challenges from a different angle. Although poor human decisions in many industrial activities have significantly impacted the environment and our sustainability, it does not mean that we are again destined to fail in our efforts. Actions based on comprehensive assessments and effective efforts can lead to starkly

different outcomes than those based on narrow views and unscientific notions. Focusing only on climate change while ignoring toxicity, however, is like caring for one's skin while neglecting the health of internal organs. Or, concerning only on energy use in cooking while omitting to thoroughly rinse pesticides and other harmful chemicals in vegetables would be analogous to such approach.

Today, we have many influential organizations such as the World Meteorological Organization (WMO), various UN departments, local and federal governments of different countries around the world, etc., actively promoting awareness and actions on climate change. However, no equivalent organization like a "World Toxicity Organization" exists to address potentially more harmful toxicity concerns comprehensively. Instead, this aspect is often downplayed by other influential bodies like the World Health Organization (WHO) and the Environmental Protection Agency (EPA), which employ irresponsible practices such as setting "allowable" or "tolerable" intake limits for harmful substances. This approach does not adequately protect humanity from these substances that accumulate and persist in nature.

Many governments have set goals through the Paris Agreement, assuming continuous human existence: carbon goals for 2030, long-term targets for 2050, and net-zero emissions between 2050 and 2100. However, the nature of toxicity accumulation challenges this assumption. Analyzing the two deadliest events in human history outside the mass extinctions from catastrophic asteroid impacts and volcanic events addresses a related context: The Black Death (Bubonic Plague, 1347-1351), with an estimated death toll of 75-200 million, caused by the bacterium

Yersinia pestis, spread through fleas on rats and devastated Europe, Asia, and North Africa and the Spanish Flu (1918-1919), with an estimated death toll of 50-100 million, resulted from interactions between human body and deadly microorganisms. Harmful chemicals share similar principles of chemically reacting in our bodies and can create much more immediate and detrimental impacts than changing weather conditions.

This reality can be readily understood through several phenomena that are currently unfolding worldwide. Clear evidence includes the high mortality rates among cotton farmers resulting from pesticide exposure, as well as the severe health threats faced by communities living near large-scale mining operations, where elevated infant mortality rates and increased incidences of cancer are well documented. Furthermore, these toxic substances spread globally through atmospheric and hydrological circulation systems, meaning that even individuals living in remote and ostensibly pristine regions now exhibit levels of toxins, such as lead and mercury in their blood, that are incomparable to those observed prior to industrialization. It is also scientifically evident that many of these toxic substances persist in the natural environment for durations that far exceed the human lifespan.

Need for Actions: These realities clearly demonstrate that toxicity must be addressed as a central issue in environmental management. I strongly believe that such an unmistakable situation should no longer be overlooked by a global environmental movement that remains narrowly fixated on carbon emissions alone. Moreover, I firmly hold that effectively resolving these concerns requires the

provision of appropriate, science-based education addressing the environment in its entirety.

Human evolution follows a clear path of continuous population growth and increased environmental impact. Ever more electronics and EVs, containing highly reactive chemical substances and consequential environmental footprints, are sold every minute globally. Environmentally harmful mining and textile production will continue to increase their outputs. Unfortunately, organizations with significant influences in our lives have catastrophically failed in effectively addressing these threatening environmental issues. As a result, there is a glaring absence of leadership to steer humanity in the right direction.

While finding ways to alleviate the toxicity burden from the ongoing human living is of critical importance, we must also recognize that there are no shortcuts or simple solutions. The challenges we face are immense, and addressing them will require a slow, deliberate process coupled with highly effective efforts. In my view, the path to a healthier environment and genuine sustainability begins with tackling the widespread misinformation that impedes progress. By empowering the readers of this book, we can inspire them to become **GREEN HEROES** in the fight for a better future.

As I publish this book, many thoughts come to my mind. Given dominant perceptions existing in our society and starkly contradictory views I presented herein, I anticipate varying reactions from readers. Despite my best efforts to provide related information grounded in science and practical experiences, I know skepticism will persist among many readers.

While I don't compare my work to history's greatest scientists, it is worth noting that Nicolaus Copernicus' heliocentric theory - that the Sun, not the Earth, is the universe's center - faced significant scrutiny. Throughout history, however, great minds have inspired others to seek truth, leading to scientific advancements and bettering our lives. Conversely, Copernicus' ideas influenced Galileo Galilei, who improved the capability of telescope technology and observed clear evidence of Earth's roundness and movement around the sun. However, Galileo faced prosecution and house arrest upon revealing the evidence despite his effective efforts.

Similarly, I anticipate significant challenges following this book's publication, albeit no house arrest :) Some may read the book skeptically from beginning to end. Others may begin with doubts and turn them into denial. Just as we still encounter flat-earthers and conspiracy theorists who reject the most convincing scientific evidence and practical experiences, such as airplanes flying over the globe and cosmic photos of Earth, it is inevitable to face similar challenges in these discussions. However, I hope that most readers evaluate the contents with open minds and, at the very least, question their priorly existing perceptions on the topics presented in the book.

In my view, these questions and doubts are healthy phenomenon. I often tell people around me not to blindly trust my words. If they care about the environment, they should invest time in research to form their own convictions rather than relying on me or any other information sources for that matter.

Unfortunately, however, this is not an easy path. Achieving adequate education in the current state of global

environmental management requires significant and effective effort, as misinformation is prevalent in our society today. It is far easier to be influenced by and echo existing perspectives than to seek truth, and many people may not voluntarily make such efforts in their free time.

While I discussed the negative outcomes of readily available misinformation on the internet, an analytical mindset can navigate this landscape to seek necessary validity from various sources. This involves earnest research with a comprehensive scope, including academic papers and reliable industry sources. If there is no evidence supporting certain claims or actions, no matter how basic they may seem, such information should not be accepted at face value.

For most readers, engaging in such thorough analysis may not be feasible. Emphasizing once again, gaining comprehensive perspectives on environmental sustainability is not about echoing random information or adopting general notions without scientific bases. Hence, **the final words of the book are dedicated for those without scientific background or time to do such investigative work:**

This book offers nearly 700 discussion points designed to help readers develop skills and analytical mindsets for evaluating information, claims, and actions in the environmental and sustainability realm. In each discussion, efforts were made to identify aspects that our society has overlooked in its pursuit of improved sustainability, grounded in scientific reasoning. Furthermore, I made several suggestions to be informed with accurate science and fight against misinformation in **Chapter V: Solutions**, which will guide readers in the right directions. Therefore, by

the very act of reading this book, readers have taken an important first step toward becoming ‘**GREEN HEROES,**’ and in doing so, they themselves become meaningful milestones guiding our society toward the right direction

Finally, you can join the “**BE GREEN HEROES (BGH)**” movement by following the **Clean Recycling Initiative™** (Non-Profit) on various social media platforms to stay informed on various topics. **It is a critical time for the GREEN HEROES to act.**

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<https://www.linkedin.com/company/78306454/admin/dashboard/>



<https://www.linkedin.com/groups/14412231/>



X:

<https://x.com/BeGreenHeroes>



Tiktok:

<https://www.tiktok.com/@begreenheroes?lang=en>



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From the Author

Thank you for reading! If you enjoyed this book, I would love to hear your thoughts. Your reviews help other readers become GREEN HEROES and support the cause. Simply click the link or scan the QR code below to leave a review on Amazon. Your feedback means the world and I read every review!

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Finally, this book is accompanied by two companion publications designed to deepen understanding of textile products, with a particular emphasis on the most material- and chemical-intensive segment: winter textiles. You can access it via the link and QR code provided below.

COLD WINTER WARM WINTER: Same Weather Different Protection – A Smart Guide to Staying Warm from Everyday Life to Advanced Textile Innovations, available globally via Amazon, including the following link and QR code:

https://www.amazon.com/COLD-WINTER-WARM-PROTECTION-INNOVATIONS/dp/B0FJY5464M/ref=tmm_pap_swat_0?encoding=UTF8&dib_tag=se&dib=eyJ2ljojMSJ9.OlliAeUIbMDZ8WTV9oq_fQ.JvAvlMTMFpw0clYUz11T0P2lPHXUBYmRXeCcQ84wdXw&qid=1753916438&sr=8-1



The AI Planet: How Artificial Intelligence Shapes – and Misguides – Environmental Narratives

- To be released soon

Document Control

Tier 3 (Educational) Document

This document forms part of the official BE GREEN HEROES (BGH) documentation framework.

Interpretation and application shall remain consistent with Tier-1 documentation where applicable.

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- End of Official BGH Document -